

Degree Project in Industrial Design Engineering

Second cycle, 30 credits

Exploring User Experience through Service Design: A Study of Mobile Applications for Public EV Charging

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Master of Science Thesis TRITA-ITM-EX 2024:557
KTH Industrial Engineering and Management
Machine Design
SE-100 44 STOCKHOLM

KTH Industriell teknik och management

Examensarbete TRITA-ITM-EX 2024:557

Utforska användarupplevelse genom tjänstedesign: En studie av mobilapplikationer för offentlig elbilsladdning

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Sammanfattning

Denna studie har undersökt användarupplevelsen av mobilapplikationer för publik laddning av elbilar. Projektet genomfördes i samarbete med företaget VCG, som är verksamt inom infrastruktur för offentlig elbilsladdning samt tillhandahåller en mobilapplikation för att hantera laddning. Syftet med detta projekt var att undersöka användarbehov i anslutning till mobilapplikationer för elbilsladdning, samt att identifiera de utmaningar och möjligheter som uppstår vid denna interaktion.

Projektet följde "double-diamond"-metoden med ett tjänstedesign-ramverk. Projektet inleddes med en litteraturstudie, där forskning om tekniken för elbilsladdning, konsumentbeteende och vanor hos elbilsförare undersöktes. Därefter genomfördes huvudstudien, där 25 användare, försäljare av elbilar och försäljare av laddboxar intervjuades, samt att observationer genomfördes på 15 olika platser. Huvudstudien följdes av ett antal olika analysmetoder som resulterade i sju huvudsakliga insikter. Dessa insikter lade grunden till den efterföljande konceptutvecklingen, där både idéutveckling och prototypskapande ingick. Konceptutvecklingen resulterade i ett huvudkoncept med flera framträdande nyckelfunktioner. Dessa nyckelfunktioner kategoriserades sedan i fem olika lösningsområden.

Projektet resulterade i en mockup av en mobilapplikation, utvecklad utifrån de olika lösningsområdena och funktionerna som identifierats i studien. Utöver mockupen presenteras tre konkreta förslag på samhällsnivå för att förbättra den övergripande användarupplevelsen.

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Abstract

This master's thesis report set out to explore the user experience of mobile applications for public EV charging. The project was conducted in collaboration with VCG, a company that operates public EV charging infrastructure and provides a mobile application to manage charging sessions. The purpose of the project was to investigate the user needs related to mobile applications for EV charging and to identify the challenges and opportunities arising from this interaction.

The project followed a double-diamond methodology, with a service design approach. The project began with a literature review exploring previous research on EV charging technology, as well as on consumer behavior and habits among EV drivers. This was followed by the main study, in which 25 users, EV retailers, and EV charger retailers were interviewed, and observations were conducted at 15 different locations. Various analysis methods were applied after the main study, resulting in seven major insights. These insights established the foundation for the subsequent concept development phase, which included ideation and prototyping. The concept development phase resulted in a main concept featuring a selection of key functions. These functions were categorized into five distinct solution areas.

The project concluded with the development of a mobile application mockup that showcases the five different solution areas and functions identified in the study. In addition to the mockup, three recommendations are presented at a societal level to enhance the overall user experience.

Acknowledgement

Recognitions and thanks.

The completion of this thesis has been a challenging yet fulfilling journey, and we want to take a moment to acknowledge those who have contributed to its success.

First and foremost, we extend our sincere gratitude to our supervisor, Sara Ilsted, for her support, thoughtful feedback, and encouragement throughout this process. Your expertise in service design has been essential in shaping this work.

We are also thankful to Mashour Afaneh from VCG. Your generosity in sharing your time and knowledge provided us with real-world insights that were crucial in grounding this thesis in practical application. We appreciate your openness and the way you welcomed us into your processes and organization.

Finally, we want to express our sincere gratitude to our friends and family. Your encouragement has kept us motivated, especially during the more difficult moments of this journey.

Batoul Al-musawy & Gustav Bennich
Stockholm, Sweden, September 2024

Nomenclature

Electric Vehicle (EV) – A type of charging vehicle that is solely powered by an electric motor.

Charging Station/Location – A location with one or more chargers for electric vehicles.

Charging Point – The outlet where the charging vehicle is plugged in. A charger can have more than one charging point.

Charging Session – The period from when the charging vehicle is plugged in and charging begins until the cable is removed from the vehicle.

Charging Time – The duration for which the charging vehicle is connected to the charging point, whether energy is being transferred to the vehicle throughout that time or not.

Fast Charging – Charging of an electric vehicle that occurs with power levels exceeding 22 kW.

Standard Charging – Charging of an electric vehicle that occurs with power levels of 22 kW or lower.

Destination Charging – Charging of electric vehicles that takes place at various types of destinations, such as commercial centers, transportation hubs, sports arenas, etc.

Charge point operator (CPO) - The entity that manages and maintains the charging stations.

Electromobility service provider (eMSP) - The entity that manages all communication and billing with EV users. Important to note that the roles of CPO and eMSP are not always distinct from one another, in some cases, one organization handles both.

Distribution system operator (DSO) - The entity responsible for operating the local electricity distribution network.

Public key infrastructure (PKI) - Is a security framework used to manage digital certificates and public-key encryption. Enables safe communication between different parties, for instance within EV charging systems.

Wireless Power Transfer (WPT) - Is the transmission of electrical energy from a power source to a device without using physical connectors or wires.

Automated Connection Devices (ACD) - Are systems or mechanisms designed to establish, manage, or disconnect electrical, mechanical, or data connections automatically.

Bidirectional Power Transfer (BPT) - Is the capability of a system to allow electrical energy to flow in both directions between a power source and a device.

Electric vehicle supply equipment (EVSE) - Regulates the transmission of electricity from the power grid to the electric vehicle.

Charging station management system (CSMS) - A platform that allows operation and control of several charging stations.

Energy management system (EMS) - A computer aided system that manages, controls, and optimizes energy usage and distribution.

Radio-frequency Identification (RFID) - uses electromagnetic fields to wirelessly read and track tags attached to items. These tags contain digital data that can be read by an RFID scanner without physical contact.

Mobile Application (APP) - Is a software program designed to perform specific tasks or provide services on devices such as smartphones, or tablets.

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1. Introduction

This chapter outlines the background of the project, the purpose and research questions to be answered, as well as delimitations.

1.1 Background

This master's thesis was conducted in collaboration with VCG, a company within the electric vehicle (EV) charging infrastructure sector. VCG operates as a public EV charging provider and offers a range of services, including public charging stations and a mobile application that allows users to manage their charging sessions. The company focuses on enhancing accessibility for EV charging through both hardware and digital solutions, such as its mobile application.

As the world transitions towards sustainable mobility, the number of electric vehicles has increased significantly in recent years, leading to a growing need for public charging infrastructure. This trend highlights the importance of intelligent solutions for public charging.

Mobile applications connected to public EV charging play a key role in streamlining the charging process by providing users with real-time information on station availability, route planning, charging costs, and payment options. Furthermore, they are important for starting and ending charging sessions, making them an integral part of the charging experience. Developing user-centered mobile applications is thus vital to enhancing the public charging experience. Additionally, organizations must align their services and business models with a user-centered focus to unlock new potential revenue streams and strengthen their market position. To this end, understanding how EV users interact with mobile applications for EV charging is essential.

This study aims to investigate EV users' interactions, needs, and challenges with mobile applications connected to EV charging, to provide insights into developing a more effective solution. Through this study, we hope to contribute to the development of a more compelling and efficient mobile application for EV charging.

1.2 Purpose and research questions

The primary objective of this research is to investigate user needs associated with mobile applications for EV charging, identifying the challenges and opportunities that emerge from this interaction. Additionally, the study aims to identify innovative, value-driven functionalities for mobile application development by analyzing user needs alongside the company's business objectives. This approach focuses on integrating user insights with organizational goals to guide the design and optimization of the mobile application, ensuring both an enhanced user experience and alignment with business objectives.

The following research questions have been formulated to help navigate the research:

RQ1: What are the EV drivers' needs for mobile applications linked to public EV charging, and how can these needs be addressed to enhance the user experience?

RQ2: How can the company enhance profitability through the mobile application functionalities while aligning with user expectations?

1.3 Delimitations

This project will focus on a Swedish context, specifically public EV chargers in the greater Stockholm area, though its findings may be applicable in other regions. The project will not emphasize the interface design (UI) of various interfaces; instead, it will concentrate on the process of developing and enhancing the quality of interaction between users and the mobile application.

2. Frame of reference

This chapter will cover EV charging technology today and customer experience while using public electric vehicle charging.

2.1 EV charging technology

Charging an electric vehicle (EV) involves transferring electrical energy from a power source to the vehicle's battery via a charging station. Charging can occur at different power levels, from slow home charging to faster public or home chargers and rapid charging stations (DC fast charging). The process is governed by standardized connectors and protocols, which ensure the safe and efficient delivery of power, along with communication between the EV and the charger. The international standard IEC 61851-1 provides a framework outlining general requirements and testing methods for EV charging systems. This standard is part of a larger set developed by the International Electrotechnical Commission (IEC) to ensure interoperability and safety within EV charging infrastructure.

As Bahrami (2020) explains, this standard establishes fundamental physical requirements for charging systems, including connectors, electrical safety, and the communication protocols between the vehicle and the charging equipment.

Connectors

In general there are four different types of connectors on the market: Type 1, Type 2, CHAdeMO, and CCS (also known as CCS Type 2), see Figure 1 for examples of the connectors.

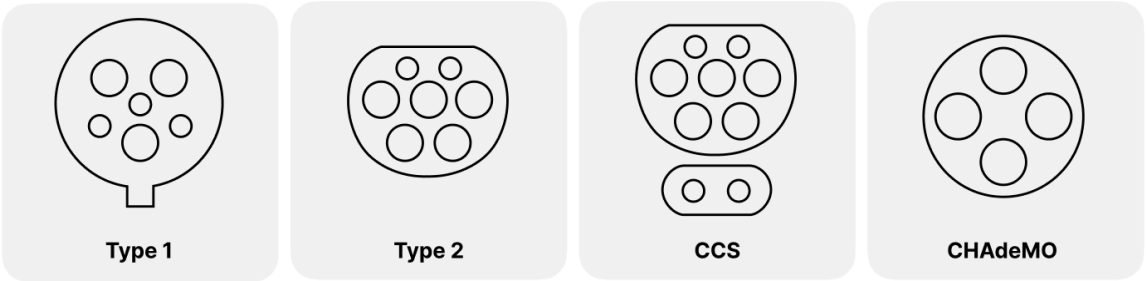


Figure 1. Illustration of Type 1, Type 2, CCS, and CHAdeMO connectors.

In Europe, it has been decided that the CCS and Type 2 connectors should be standardized, which means that cars from European manufacturers are typically equipped with these connectors.

The standard establishes a framework for four different charging modes, specifying different charging levels and safety features. The modes are: Mode 1, Mode 2, Mode 3, and Mode 4. For public charging, the EU has decided that Mode 3 should be the standard for normal and semi-fast charging, while Mode 4 should be the standard for fast charging. Below is a more detailed description of the different modes of the IEC 61851-1 standard.

Mode 1

The lowest safety level in electric vehicle charging. Mode 1 involves charging from a standard grounded wall outlet, providing 230/400V at up to 16A. The charging process occurs entirely without monitoring.

Mode 2

This safety level involves, similar to Mode 1, charging from a standard grounded wall outlet. However, in this level, the charging process is monitored using a safety box mounted on the charging cable. This safety box contains, among other things, a residual-current device and components that can communicate with the car to ensure controlled charging.

Mode 3

The European safety standard for normal and semi-fast charging since 2017. This safety level ensures that the car and charging station communicate before charging begins. Therefore, the charging cable remains unpowered until all tests are completed, allowing charging to occur entirely without risk.

Mode 4

Is the European standard for direct current (DC) charging, which supports fast charging. These charging stations are often found along roads where quick charging is necessary. A high safety class is required because charging occurs at very high power levels.

Actors in the network

In the charging network at charging stations, several key actors collaborate to ensure efficient and reliable service for EV drivers. These actors include the EV Driver, Charging Point Operator (CPO), E-Mobility Service Provider (eMSP), Electricity Supplier, and Electricity Distribution Company.

- The **EV Driver** is the end user who interacts directly with the charging station to recharge their EV.
- The **Charging Point Operator (CPO)** manages and maintains the physical infrastructure of the charging stations. They ensure that charging points are operational, monitor their performance, and handle maintenance and technical support.
- The **E-Mobility Service Provider (eMSP)** offers services to EV drivers, such as mobile apps, charging subscriptions, or payment solutions. The eMSP acts as an intermediary between the EV driver and the CPO, often handling the billing and user interface for accessing multiple charging stations across different networks.
- The **Electricity Supplier** provides the energy needed to power the charging stations. They supply electricity based on agreements with CPOs, ensuring the charging stations receive the required energy to operate.
- The **Electricity Distribution Company** is responsible for delivering electricity from the grid to the charging stations. They manage the local infrastructure, ensuring stable and reliable distribution of electricity from the supplier to the charging stations.

These actors work in an interconnected system: the EV driver relies on the eMSP for access and payment services, while the CPO operates the physical charging points and ensures connectivity. The Electricity Supplier and Electricity Distribution Company collaborate to provide the essential power to the charging network, enabling a smooth charging process for all parties involved. Figure 2 shows the actors involved and how they are related.

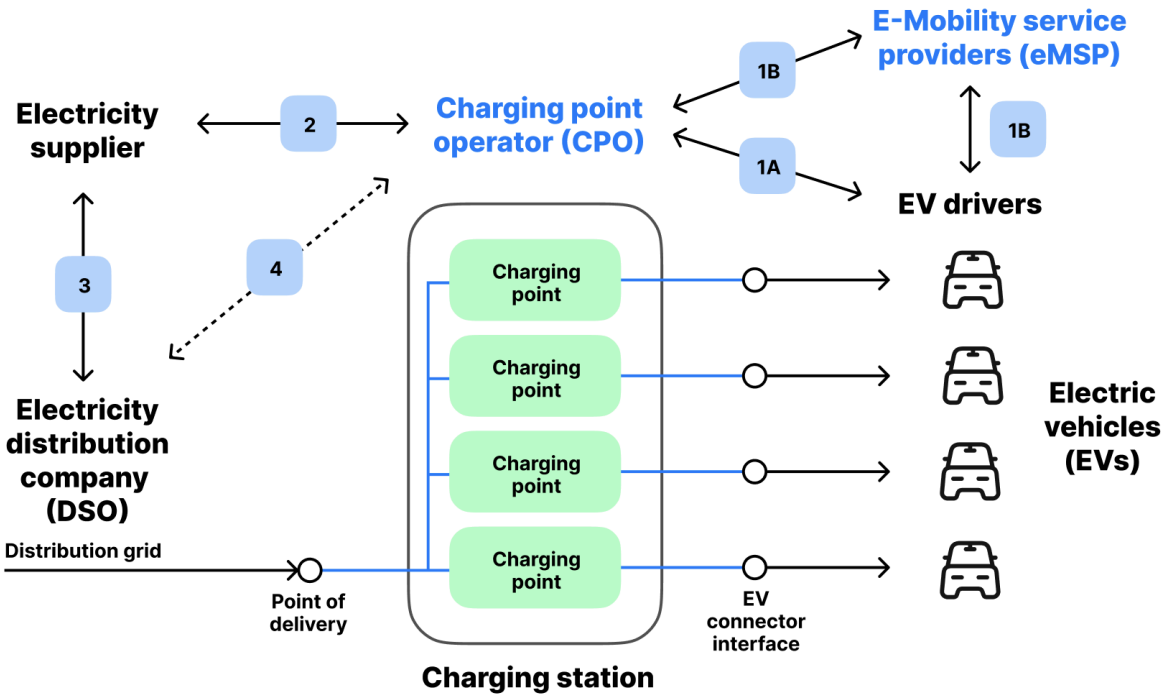


Figure 2. Distribution of main actors in the charging network at a charging station, and their relation.

Legend: 1A: EV Driver - Charging point operator (CPO) (directly), 1B: EV Driver - Charging point operator (CPO) (intermediated by E-Mobility service providers (eMSP)), 2: Charging point operator (CPO) - Electricity supplier, 3: Electricity supplier - Electricity distribution company, 4: Electricity distribution company - Charging point operator (CPO) (only for connections).

2.2 Smart Charging

The accessibility of charging stations is essential for the widespread adoption of electric vehicles. The electrification of transportation has been acknowledged as one of the main factors contributing to the rise in power demand. Nonetheless, the electricity grid faces challenges from uncoordinated charging, which can lead to power losses, overload, and a decline in grid dependability indices (Deb et al., 2022).

According to Wang et al. (2016), the growing number of EVs could present significant challenges to the grid. For example, the load incurred when an EV is charged at level 2 alternating current (AC) charging power can be twenty times greater than what is needed to support a typical North American home.

Smart charging emerges as a solution to a range of challenges regarding the power grid, and according to Deb et al. (2022), smart charging enables a certain degree of regulation over the charging procedures. Therefore, the adaptability of smart charging in the charging process is essential for meeting the requirements and needs of both the power system and the EV user. The use of smart chargers enables the charging power to be dynamically adjusted based on the available power from the grid, the specific needs of EV users, and also provides emergency grid backup. Table 1 presents a detailed explanation for the four types of smart charging technologies.

Table 1. The four types of smart charging, including the control approach and its applications (Deb et al., 2022), (Anisie et al., 2019), (Werneskog & Wolf, 2023).

Forms of smart charging	**Type**	**Control approach**	**Use**
Uncontrolled	Uncontrolled charging without automated control (with time-of-use pricing)	None	Grid capacity management over the long run via for example peak shaving.
Basic Control	Basic charging control	Controlling the charging process with On/Off mechanism.	Grid congestion control
Unidirectional	V1G (Unidirectional controlled charging)	The charging rate can be changed in order to rapidly respond to the grid conditions.	Frequency control and additional services.
Bidirectional V2X (Vehicle-to-Everything)	V2G (Vehicle-to-Grid)	Aims to the ability to output power from the EV to the surrounding systems. Enables the EV in the discharge mode to deliver services to the grid. Another form of bidirectional charging is V2H/B that allows the EV's to deliver supplement or back-up power to homes and buildings when there is power blackout or to increase the amount of energy generated locally. V2V and V2L refers to the ability of the EV to charge another load or vehicle without using the grid.	Short-term adoption of green power, additional services, frequency control, voltage control and load after an increase in reliability.
	V2H/B (Vehicle-to-Home/ building) Also known as V2C (Vehicle-to-customer)		
	V2V (Vehicle-to-Vehicle)		Optimization of the microgrid.
	V2L (Vehicle-to-Load)		

One of the four forms of smart charging is bidirectional charging, also known as Vehicle-to-Everything (V2X), a broad term that encompasses several methods of supplying power from the EV to external equipment (Werneskog & Wolf, 2023). The different varieties of V2X are shown in Figure 3.

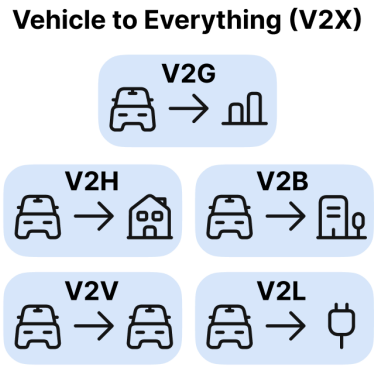


Figure 3. Varieties of vehicle to everything (V2X).

According to research conducted by Khezri et al. (2022), using V2X technology, EVs could aid in reducing electricity costs, frequency control, voltage control, and load management, thereby increasing reliability. Figure 4 shows some of the benefits of different applications of V2X.

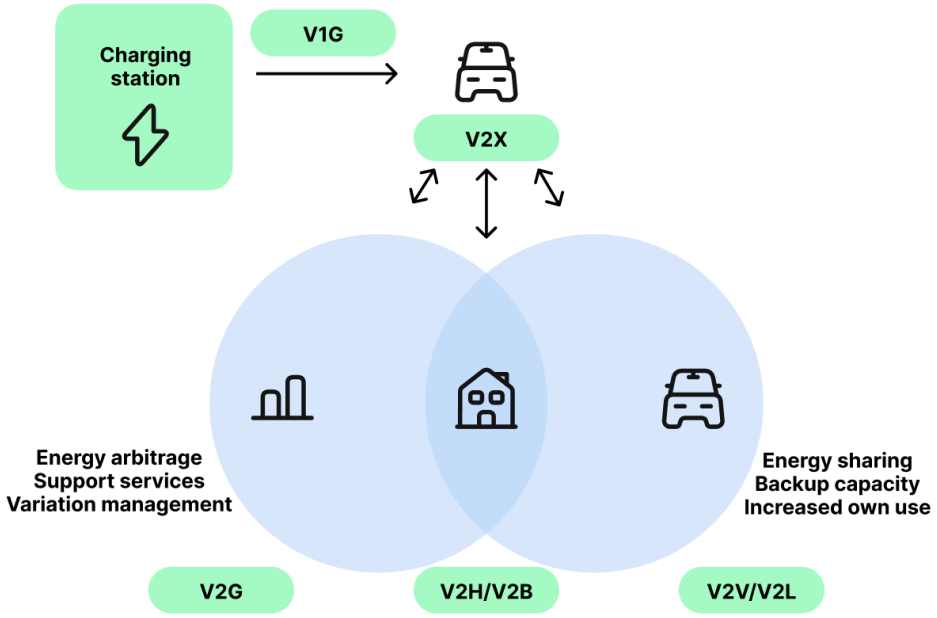


Figure 4. Examples of the benefits and applications of V2X.

2.2.1 Economic benefits of smart charging

The growing adoption of smart charging offers substantial economic benefits by optimizing energy usage and reducing costs associated with the increasing demand from EVs (Chen & Zhang, 2024). Anisie et al. (2019) state that stacking various revenue streams presents a significant opportunity for enhancing both system and local flexibility. Services like buying and selling electricity at different times, frequency reserves, voltage control, and congestion management via peak shaving are examples of these revenue streams, each offering distinct financial opportunities.

EV drivers also benefit from smart charging, particularly with V2G technology. By participating in V2G services, EV owners can reduce their charging costs and potentially earn profits by charging during low-tariff hours and utilizing stored electricity. Additionally, by engaging in energy arbitrage on behalf of users, aggregators can generate profits and gain benefits (Chen & Zhang, 2024).

A study by Visaria et al. (2022) shows that EV owners choose EVs primarily to reduce operating costs. Furthermore, potential savings from lower per-kWh charging costs are significant enough that some owners are willing to pay a premium for the advantage of cheaper everyday charging. Respondents indicated a willingness to pay a subscription fee of 33.4 DKK/month (approximately 52 SEK/month with an exchange rate of 1 DKK to 1.56 SEK) for each 1 DKK/kWh reduction in public charging costs.

In brief, smart charging technologies offer significant economic advantages for both operators and EV users by optimizing energy use and reducing costs. EV users benefit from charging during low-tariff periods, and studies highlight the financial motivations for this shift, showing that EV users prioritize lowering operating costs and are often willing to pay for reduced public charging costs.

2.3 Communication protocols

Standards and protocols play a key role in fostering interoperability not only between various actors, systems, and products, but also with the surrounding infrastructure. European regulations have so far focused on standardizing hardware, for example by mandating the use of Type 2 connectors for alternating current and CCS connectors for direct current charging (Werneskog & Wolf, 2023).

The electromobility and energy sectors are becoming more interconnected, encompassing a wider range of functions in the electromobility ecosystem as smart and bidirectional charging become more prevalent. To support the system and enable the benefits of smart charging, communication protocols need to be kept up-to-date to reflect changes in capabilities and functionalities (Meersmans, 2023).

Various communication protocols are employed at different stages of the communication chain; some of these have specific smart and bidirectional features and the ability to achieve the intended ecosystem results (Meersmans, 2023). The role of these protocols is to facilitate the exchange of data, such as charge point locations and authorizations, as well as to send commands for charging control. Four primary channels of communication have been identified (Klapwijk & Driessen, 2017). Table 2 provides a description of these four different communication channels, along with the various protocols used.

Table 2. Different communication channels, description and protocols (Klapwijk & Driessen, 2017), (Wellisch et al., 2015).

Communication channel	**Description**	**Protocols**
EV - EVSE	Communication between the vehicle and the charging station.	IEC 61851-1, ISO 15118
EVSE- CPO	Communication between the charging station and the CPO	OCPP, IEC 61850-90-8
CPO - eMSP	Communication between CPO and eMSP, also known as roaming in the EV field. Roaming enables the driver to use the same identification to charge outside of their CPO's charging network.	OCHP, OCPI, OICP, eMIP
CPO - DSO	Communication between the CPO and actors in the grid system.	OSCP, IEC 62746-10-1 (Open ADR 2.0)

An illustration of the communication channels and their relation to the different actors is shown in Figure 5. A complete illustration of the various protocols between the different actors can be seen in Appendix A.

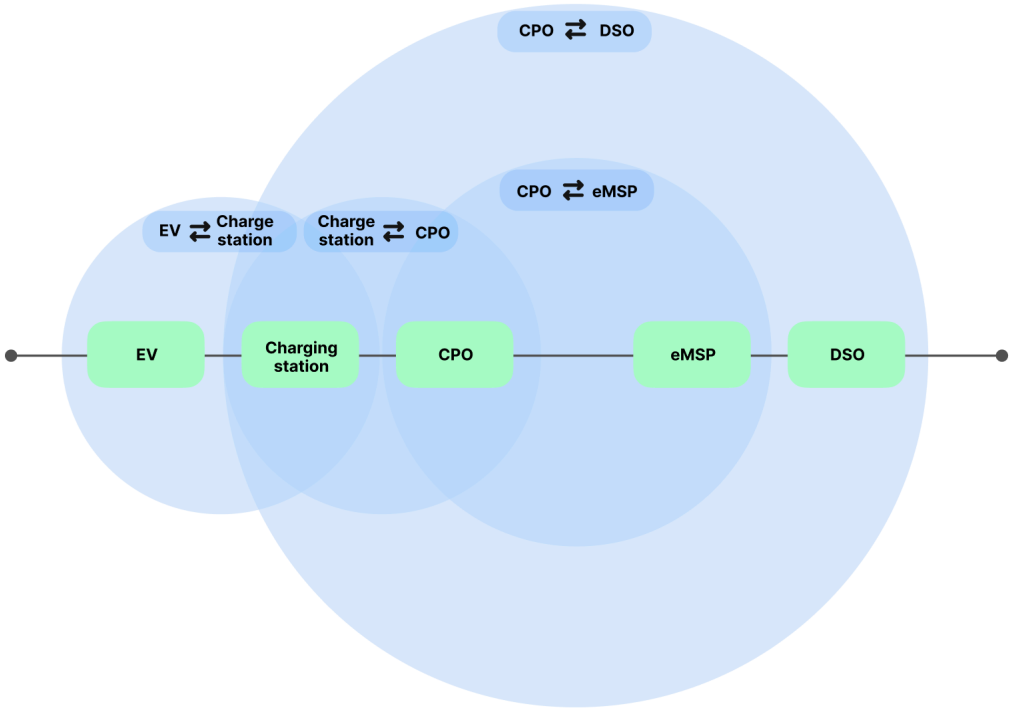


Figure 5. An illustration of the communication channels and the actors involved.

For this project, two of the protocols are investigated further: ISO 15118 and OCPP. These protocols directly affect the project, the EV users, and the EV charging provider, who are the primary stakeholders. The other protocols primarily influence the grid and charging system, rather than the actual user experience.

2.3.1 ISO 15118

ISO 15118 is an international communication protocol standard, developed through joint work between ISO and IEC. This standard specifies how charging stations and electric vehicles should communicate with each other and enables the use of information and communication technologies to integrate EVs into the smart grid and support more advanced features (Multin, 2018). Establishing a more advanced and self-operating charge control system between electric vehicles and charging infrastructure is the main goal of ISO 15118 (Meersmans, 2023).

To date, there have been two released versions of the ISO 15118 protocol. In 2015, the initial version of this standard, ISO 15118-2, was released. A more recent version, ISO 15118-20, was released in 2022. Notably, ISO 15118-20's new message structure prevents it from being backward-compatible with ISO 15118-2. Table 3 presents a comparative overview of the two

versions of ISO 15118, highlighting the supported charging technologies for each version. Compared to ISO 15118-2, ISO 15118-20 adds support for Wireless Power Transfer (WPT), Automated Connection Devices (ACD), and Bidirectional Power Transfer (BPT) (Meersmans, 2023).

Table 3. Comparison of ISO 15118-2 and ISO 151118-20 regarding supported charging technologies (Meersmans, 2023).

ISO 15118 protocol version:	**AC** Alternating current	**DC** Direct current	**WPT** Wireless Power Transfer	**ACD** Automated Connection Device	**BPT** Bidirectional Power Transfer
ISO 15118-2	✓	✓			
ISO 15118-20	✓	✓	✓	✓	✓

ISO 15118 introduces a multitude of features to the electricity and electromobility ecosystem. Among these advancements, the most critical and relevant are:

Extension of supported charging technologies

While the primary focus of ISO 15118 is on EV charging, it is applicable to other types of vehicles as well. The standard encompasses wired (AC and DC), Wireless Power Transfer (WPT), and Automatic Connection Devices, including pantographs (ACD), as part of the Combined Charging System (CCS). It further supports Bidirectional Power Transfer (BPT), which enables the integration of EVs into the smart grid (Meersmans, 2023).

EV Authentication Mechanisms

Plug & Charge (PnC) is a feature of the ISO 15118 standard and allows the EV to authenticate with the charging station on behalf of the driver (Meersmans, 2023). Bao et al. (2017) explains in their research that PnC automates authentication by using digital certificates provided by public key infrastructure (PKI) for authorization and authentication. For PnC, an EV's authenticity and association with a specific mobility contract between the EV owner and the mobility operator are determined by contract certifications. This process eliminates the need for external identification methods, such as RFID cards, which simplifies the charging experience.

Varied control modes

According to Meersmans (2023), ISO 15118-20 supports two control modes, scheduled and dynamic, which determine who controls the charging schedule.

In *scheduled control mode*, the EV determines the charging schedule to be followed, based on the driver's mobility needs. Only restrictions on the charging schedule that take local grid limitations into account can be communicated by the electric vehicle supply equipment (EVSE). To ensure that the charging schedule continuously respects grid infrastructure limits, the process by which the electric vehicle determines the appropriate charging schedule is subject to re-negotiation. This ensures that the electric vehicle will be charged within the boundaries of the local grid.

When the charging station has control over the charging session, it is called *dynamic control mode*, which was introduced in the updated version of the protocol, ISO 15118-20. The EV communicates its mobility needs to the charging station, allowing it or any external actor, such as the CPO or energy management system (EMS), to determine an appropriate charging schedule that meets the driver's preferred level of comfort. Additionally, dynamic control mode allows the charging station or other external actors to send updated mobility needs to the electric vehicle if the driver modifies their preferences via the user interface. This mode also enables energy calculation, based on the current mobility needs, the electric vehicle can adjust its intended energy levels, such as minimum energy and targeted level of energy.

Support for data exchange of battery state of charge (SoC)

ISO 15118-20 enables the optional exchange of battery-related information, including the current SoC and battery capacity, along with the mobility needs in kWh, from the electric vehicle to the charging station , allowing drivers to access this information from sources other than the car's infotainment interface (Meersmans, 2023).

2.3.2 OCPP

The Open Charge Point Protocol (OCPP) is a protocol intended to standardize communication between the charging station and a central system that manages and operates the charge points. The OCPP protocol enables the Charging Station Management System (CSMS) of the CPO and the charging station to communicate using the OCPP protocol (Klapwijk & Driessen, 2017). Figure 6 illustrates the OCPP interaction between the charging station and central system.



Figure 6. OCPP interaction (Wellisch et al., 2015).

Research conducted by Klapwijk and Driessen (2017) explains that OCPP enables the possibility of switching charging networks without needing to replace the charging stations or perform extensive reprogramming, due to its open and widely available nature. Meersmans (2023) explains further that EVSE owners can easily switch to another OCPP-based system if, for example, the manufacturer of the EVSE goes out of business. Additionally, the ability to freely switch to any OCPP-based system provides users with the flexibility to use any network or charging station. This promotes free competition among different manufacturers of charging stations and networks, leading to more competitive pricing, improved quality, and increased innovation. The OCPP protocol offers a multitude of functions, such as authorizing charging sessions, handling billing, managing the grid, operating charge points, making reservations, and enabling smart charging (Klapwijk & Driessen, 2017).

Meersmans et al. (2023) moves on to explain that there are several versions of this protocol, but the dominant version among charging stations and CSMS platforms is OCPP 1.6, which was released in 2015. However, newer versions have since been released, such as OCPP 2.0 in 2018, OCPP 2.0.1 in 2020 and the latest version, OCPP 2.1, is in progress and is expected to be released. Following the release of version 2.0, the Open Charge Alliance issued version 2.0.1 to address various protocol challenges. An overview of the various versions of the OCPP protocol is provided below:

OCPP 1.6

In contrast to earlier versions, OCPP 1.6 is the first version to offer some support for smart charging. Meersmans (2023) explains that the protocol includes a module called “smart charging”, which introduces the ChargingProfile. The ChargingProfile is a tool for managing how and when an EVSE charges an EV. It includes a charging schedule that defines start and end times, as well as the maximum power or current allowed at different times. Only the central system can send a ChargingProfile to the charging station, and a current, active ChargingProfile can also only be cleared by the central system. The ChargingProfile has two

main attributes: ChargingProfilePurpose and ChargingProfileKind, as explained by Meersmans (2023):

ChargingProfilePurpose refers to the objective of the charging profile and offers three options. The first is ChargingStationMaxProfile, which determines the maximum power or current that each charging point in a station may utilize. The second option, DefaultProfile, provides standardized charging schedules that are typically used to enforce charging policies. Finally, TxProfile is a transaction-based charging profile that can dominate the DefaultProfile only while the transaction is active.

ChargingProfileKind determines when the charging schedule should be applied and includes three options: “absolute,” used for a fixed start time, ideal for schedules aligned with dynamic time-of-use prices; “recurring,” for schedules that restart regularly, like daily, to align with static time-of-use tariffs; and “relative,” for immediate activation when rapid charging is needed.

In summary, OCPP 1.6's smart charging functions primarily focus on different levels of scheduled charging capabilities. The protocol offers more flexibility in managing charging based on user needs, time-of-use tariffs, and demand management.

OCPP 2.0.1

Version 2.0.1 of the OCPP protocol introduces new and updated features designed to better reflect and meet the needs of the ecosystem. One of the more prominent features is increased accommodation for driver needs, as explained by Meersmans (2023):

The previous version of the protocol, OCPP 1.6, introduced some support for smart charging but does not support the ISO 15118-2 communication protocol. Due to this lack of support, critical information from the EV, such as energy requirements and the driver's departure time, can not be transmitted to the charging station and the CSMS. OCPP version 2.0.1 adds two new messages that allow information from the EV to be shared, enabling more customized charging according to the requirements of the driver. This allows the CSMS to propose and calculate a charging schedule that fits the EV's needs. As a result, the EV can either follow the suggested schedule or create its own charging schedule based on the proposed plan, as long as the plan is within specific bounds.

OCPP 2.1

Version 2.1 of the OCPP protocol is expected to enable bidirectional charging and support the latest protocol version, ISO 15118-20. Several new features expected to be introduced are outlined below, as anticipated by Meersmans (2023):

The first feature is *increased support for V2X operation modes*. New modes aimed at supporting and managing bidirectional charging are expected. In the latest version of the ISO 15118-20 communication protocol, an external actor, not just the EV, can be responsible for calculating and forecasting a dynamic charging schedule. This addition enables the charging station to receive information from the CSMS regarding bidirectional charging processes.

The second feature is *added support for priority charging*. A new feature called "PriorityCharging" is expected to be introduced in version 2.1 of the OCPP protocol. This feature has been added to ChargingProfilePurpose and allows the EV driver to bypass various smart charging options to charge as quickly as possible. This is particularly advantageous when the EV driver wants to prioritize speed over smart charging strategies.

2.4 EV consumer behavior and habits

This section explores the charging behaviors and preferences of EV users, providing insights into the factors that influence their decision-making and charging habits.

2.4.1 EV consumer overview: behavior and experience

Understanding consumer behavior and optimizing charging infrastructure to improve the overall user experience is essential. A study by Busengdal et al. (2022) on consumer behavior among EV drivers reveals that data from Norway indicates EV drivers are predominantly men aged 45 to 66, with 73% living in detached homes and having access to a private parking space. Furthermore, a study by Visaria et al. (2022) indicates that most EV users tend to be higher-income earners with higher levels of educational attainment. Additionally, the survey shows that 80% of Norwegian EV owners have a home charger installed, with 23% charging daily, 31% every two to three days, and 25% charging once a week (Busengdal et al., 2022).

Moreover, Fischer et al. (2024) state that half of all charging sessions occur at home, a trend attributed to common characteristics shared by most EV owners. Globally, 48% of all EV owners have access to some form of home charging, and an additional 35% could charge at

home with certain upgrades, such as a network upgrade or the installation of an extension cable. Only 17% of EV owners lack the possibility to charge at home.

Building on Fischer et al. (2024), who highlight the dominance of home charging, Lindblad et al. (2023) provide additional insights specific to Stockholm, Sweden. According to data from 2022, the distribution of public charging sessions in Stockholm shows that 47% of all sessions involve regular AC charging in parking garages, 32% occur through regular AC charging on public streets, and the remaining 21% involve fast charging on public streets. The study also offers insights from 2022 on the daily distribution of charging sessions. Data indicate that most sessions occur on weekdays, with the highest number starting between 07:00 and 10:00, peaking between 08:00 and 09:00 for regular AC charging in parking garages, primarily used for destination charging. In contrast, fast charging sessions are most common midday, peaking between 13:00 and 14:00. Moreover, data from the City of Stockholm show that most charging sessions last less than three hours, a pattern explained by local time restrictions in Stockholm, with a 3-hour limit for regular charging and a 30-minute limit for fast charging (Lindblad et al., 2023).

Consumer experiences reveal dissatisfaction with public charging infrastructure, as evidenced by data from France. A self-reported survey of EV owners highlights the reasons for this dissatisfaction. According to the study, 50% of respondents cite unclear pricing, and 38% report that charging points are frequently out of service. Other reasons for dissatisfaction include non-charging vehicles occupying charging spots, poor service, and limited payment options, as shown in Figure 7 (Statista, 2024).

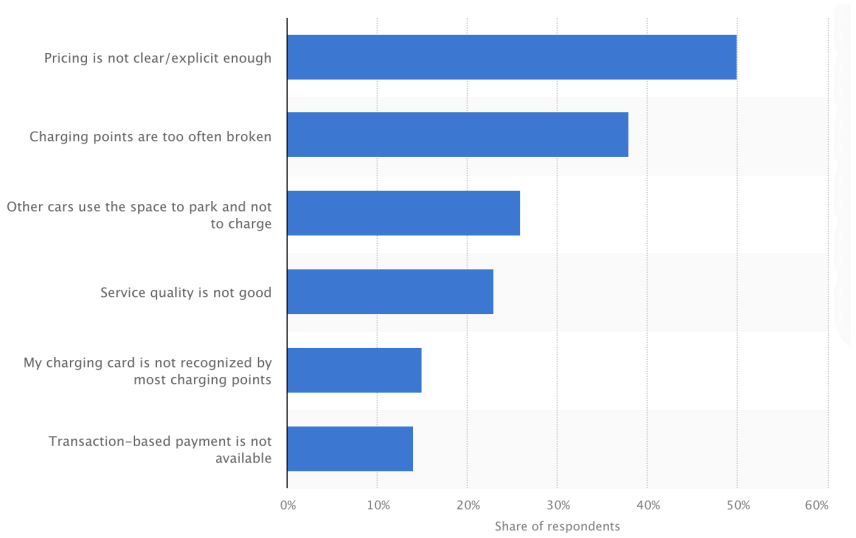


Figure 7. EV consumer dissatisfaction reasons and percentage of respondents (Statista, 2024).

In addition, a study conducted by Busengdal et al. (2022) in the Nordic countries reports the barriers non-EV drivers face in purchasing an EV. A significant number of survey participants identified the price of the car as a barrier, and in Sweden and Denmark, the second most significant barrier is the lack of charging opportunities at home or nearby.

In conclusion, research shows that home charging is the most popular option for EV owners worldwide, with regional variations in charging behavior and satisfaction. However, issues with public charging infrastructure, such as unclear pricing and unreliable charging points, continue to affect overall consumer satisfaction.

2.4.2 Consumer preferences during public charge point selection

The increasing number of EVs has underscored the need for robust charging infrastructure, highlighting the importance of a positive customer experience. A substantial body of literature examines the multifaceted challenges and opportunities within this domain..

One of the primary challenges is the inconsistent availability and accessibility of charging stations. Research by Neaimeh et al. (2017) highlights that charging station scarcity, especially in rural areas, leads to "range anxiety," deterring potential EV adopters. Furthermore, charging speed is a critical issue. Slow charging times, particularly at Level 2 chargers, are perceived as a major inconvenience compared to the rapid refueling of conventional vehicles (Morrow et al., 2008). Interoperability and payment system complexities also pose significant barriers. Hardman et al. (2016) note that the lack of standardized connectors and payment methods across different networks frustrates users and complicates the charging process.

In this context, the choice of public charging stations plays a significant role in user experience and satisfaction. According to a study by Fischer et al. (2024), survey data indicate that 42% of respondents ranked charging speed as the most important factor, highlighting its significance in decision-making. Additionally, consumers expressed a preference for charging times of no more than 30 minutes, emphasizing the need for fast and efficient charging solutions to meet customer demands. Beyond charging speed, cost is considered the second most important factor when selecting a public charging station. However, the application of the charging station influences how much people are willing to pay; for example, EV drivers are generally willing to pay 10% more for highway charging

compared to inner-city or destination charging. Figure 8 provides a graph of key factors influencing consumer preferences when selecting public charging points, based on the study by Fischer et al. (2024).

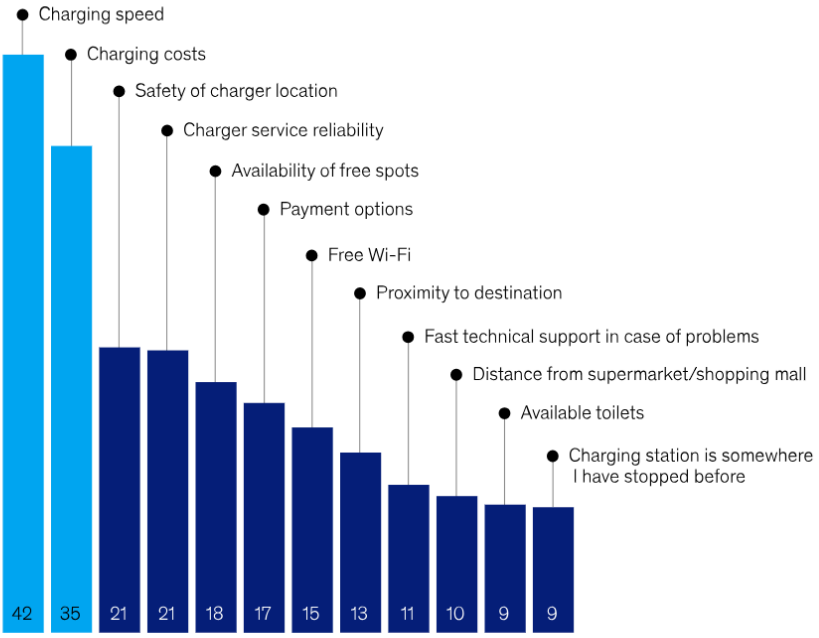


Figure 8. Graph of key factors influencing consumer preferences when selecting public charging points (Fischer et al., 2024).

Conversely, advancements in fast-charging technologies and the expansion of charging networks present significant opportunities to enhance the EV customer experience. The deployment of ultra-fast chargers, as discussed by Coffman et al. (2017), may drastically reduce charging times, improving convenience for EV users during long-distance travel. Additionally, integrating smart grid technology can optimize charging patterns and reduce peak demand, as explored by Richardson (2013). This not only enhances efficiency but also lowers costs for consumers. Furthermore, the rise of app-based platforms for locating charging stations streamlines the user experience by providing real-time data and increasing convenience (Mak et al., 2013).

A study by Rastad et al. (2023), conducted across seven European countries, highlights the diverse motivations for using smart charging among EV drivers. The most commonly cited reason is cost savings, which is the primary motivator for smart charging. Furthermore, a significant portion of respondents prioritize using greener energy as their main motivation for adopting smart charging practices.

In conclusion, although the current EV charging infrastructure presents several challenges, such as availability, charging speed, and interoperability, advancements in technology and network expansion provide promising opportunities. Addressing these challenges with innovative solutions is crucial for enhancing the customer experience and accelerating EV adoption.

3. Methodology

This chapter provides an overview of the research design employed in this project. It includes an explanation of the design thinking methodology, followed by a detailed description of the stages and procedures implemented.

3.1 Research design and approach

The methodology used in this study is based on the design thinking process. Design thinking focuses on addressing innovation and design from a human-centered approach. This process aims to foster creativity and supports the development of successful business models. This approach combines the methods used by designers for generating concepts, designs, and innovations with a methodical, fact-based evaluation of the economic feasibility of these ideas (Muller-Roterberg, 2020).

The design thinking process, based on the double diamond framework, consists of four key stages: Discover, Define, Develop, and Deliver. This approach streamlines the design workflow and facilitates effective communication between designers and non-designers, which is essential due to the complexity of design challenges. Additionally, it provides a framework for evaluating project progress, especially in cases where outcomes are uncertain, and emphasizes key stages while maintaining flexibility (Kochanowska et al., 2021).

This framework combines a user-centered approach to innovation with the flexibility, resources, and tools necessary to navigate the complexity and dynamics of the research. The process was iterative, adhering to the Double Diamond framework, which included multiple rounds of research, interviews, and testing. Figure 9 illustrates the different phases of the Double Diamond framework and the methods used within each phase.

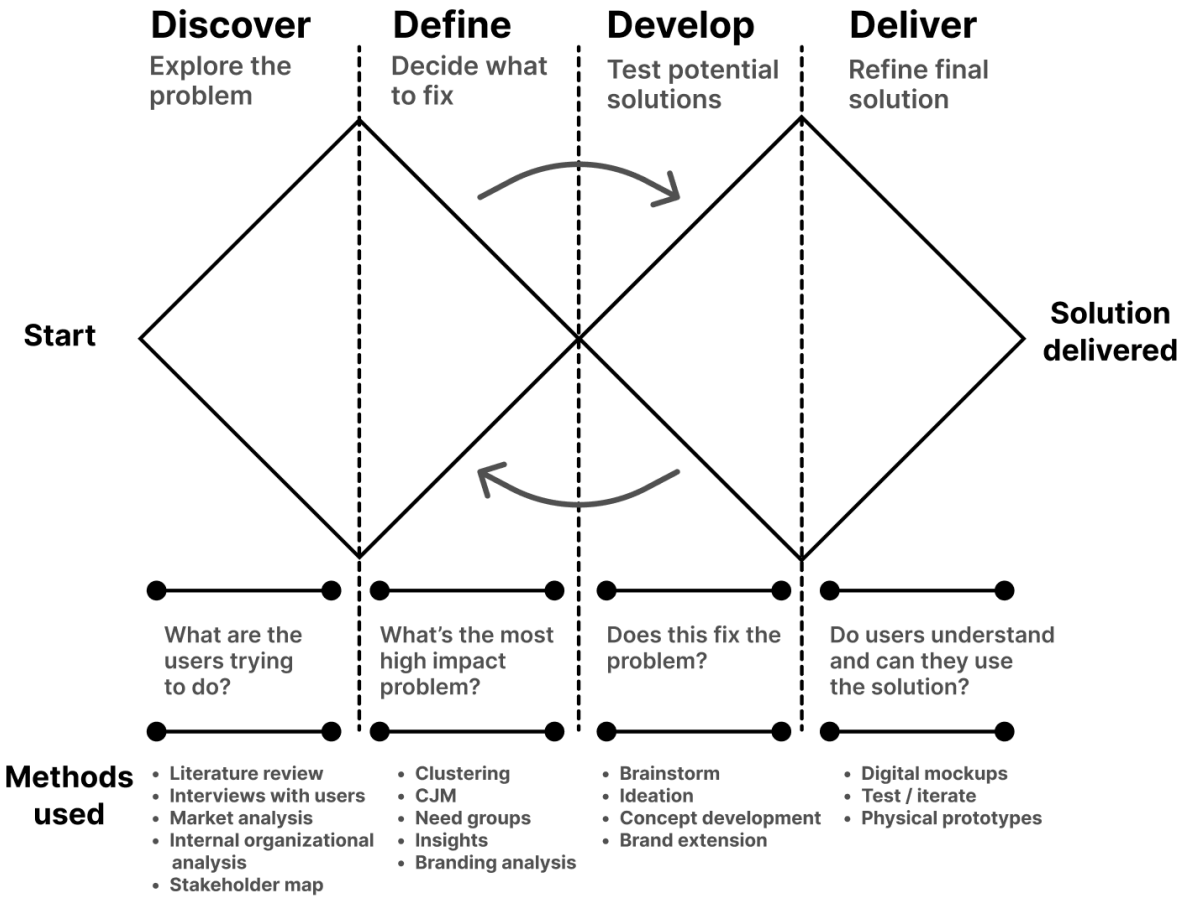


Figure 9. Visualization of the double diamond approach and methods used under each stage of the double diamond.

The **Discover** phase focused on gathering information to understand and analyze the problem. This phase aimed to broaden the perspective and address both what the problem is and why it exists. A significant part of this phase involved understanding users, their needs, and the underlying reasons for those needs. A user analysis was conducted, including observations and interviews. The observations focused on EV users' interactions with mobile applications in real-life situations to gain insights into their behaviors and identify areas for improvement. Interviews were conducted with EV users to obtain detailed insights into their experiences, needs, preferences, and pain points.

The Discover phase included a thoroughly conducted in-depth literature review to address information gaps and incorporate up-to-date research. This literature study also aimed to evaluate existing theories and develop new ones through empirical data, in conjunction with the user analysis conducted for this study. Additionally, it was essential to understand the entire stakeholder network along with trends and technical solutions in the field. This

knowledge helped identify what would be feasible and how to meet users' needs within the mobile application. Given that this is a service design project integrated with business modeling, it was crucial to grasp the overall picture and balance customer needs with the business needs of the company involved.

Additionally, a comprehensive market analysis was conducted, mapping 11 of the largest electric charging providers in Sweden, with a detailed comparison of all their mobile application features. The purpose of this analysis was to gain insights into emerging trends within the field and closely examine the business model's environment, particularly the industry forces. By analyzing certain aspects of the business model, such as the value proposition, customer segments, and revenue streams according to Osterwalder and Pigneur (2013), a comprehensive understanding of its surrounding environment became essential for developing more robust and competitive business models.

The **Define** phase centered on defining the challenge and target users based on the insights gathered from the Discover phase. The insights from the Discover phase were analyzed, filtered, and ultimately refined into needs and several problem areas. These needs were then used to compile the findings into key insights. To analyze, organize, and incorporate the insights acquired during the Discover phase, the clustering technique was applied. Responses were grouped into sections based on similar themes and response types to identify patterns.

Customer Journey Mapping (CJM) was conducted to visualize the user's journey from start to finish within EV charging mobile applications. This approach helped identify each step in the user experience. By analyzing these stages from a user-centric perspective, potential areas for improvement were identified and addressed effectively.

In conclusion, the first two phases, Discover and Define, guided by the framework's diverging and converging thinking, focused on the 'what' and 'why,' emphasizing the importance of utilizing user insights to drive effective business development in the face of evolving market dynamics. This approach ensures that the solutions not only address user needs but also align with broader business objectives and market opportunities.

During the **Develop** phase, the primary objectives were to generate, test, and refine potential solutions. The ideation process began with team brainstorming, resulting in a wide range of ideas aimed at addressing various aspects of the problem. A program called *Figma*, a cloud-based digital whiteboard and prototyping tool, was used as an open canvas, enabling

the organization and clustering of ideas, which then served as the foundation for rapid prototyping.

The **Deliver** phase involved the finalization and implementation of the solutions developed during the Develop phase. During this stage, digital mockups were created to visually and functionally represent the refined concepts. Throughout this stage, iterative refinement and continuous testing were essential for improving the digital mockup. In addition to the mockup, a new brand extension was developed to align with the new service extension. Along with the actual mockup, broader insights from the earlier stages of the project, which extend beyond the mockup's scope, were formulated.

3.2 Service Design

This project adopts a service design approach, which encompasses a broad and multifaceted definition. Service designers often interpret the term in various ways (Stickdorn et al., 2018). Stefan Moritz provides a well-known definition that aptly captures the essence of this project:

> "*Service design helps to innovate (create new) or improve (existing) services to make them more useful, usable, desirable for clients as well as effective for organizations. It is a new holistic, multidisciplinary, integrative field* " - Stefan Moritz (Stickdorn et al., 2018).

Further**,** the principles of service design are explained by Stickdorn et al. (2018) and Stickdorn and Schneider (2012):

User-centered: Service design prioritizes the user, taking into account the experiences of all stakeholders. A service is created through the interaction between provider and customer, with customer participation essential to its success

Collaborative/Co-creative: In the service design process, it is essential to consider and engage all stakeholders collaboratively. Collaboration with various stakeholders, including different customer groups, employees, managers, and non-human interfaces, is essential in exploring and defining the service proposition.

Sequencing: The service design process is viewed as dynamic, with ongoing processes composed of connected actions. Therefore, while designing the service, it is essential to consider the timeline to ensure a positive overall customer experience. A well-paced service flow minimizes stress and dissatisfaction caused by delays and rushed tempos.

Iterative: Service design follows an approach focused on exploration, adaptability, and experimentation, including continuous iteration, concept testing, and refining. Iteration ensures improved service effectiveness.

Real: Service design emphasizes realistic process grounding, where user needs, for instance, should be researched in real-life contexts. The ideas and prototypes should also be tested in practical settings.

Holistic: In the service design approach, the broader context in which the service operates is considered. The needs of all stakeholders should be addressed, including the value chain of the entire service and the business as a whole.

4. Main study

This chapter presents the main study, including user analysis, market analysis, and stakeholder mapping. The user analysis included observations and interviews with users, as well as with EV charging retailers and EV retailers.

4.1 Observations

The initial observations were conducted on weekdays across the greater Stockholm area to learn how the physical charging locations looked and how the stations differed from each other. Normal and fast chargers were observed at various locations. The charging locations could be categorized as presented in Figure 10.

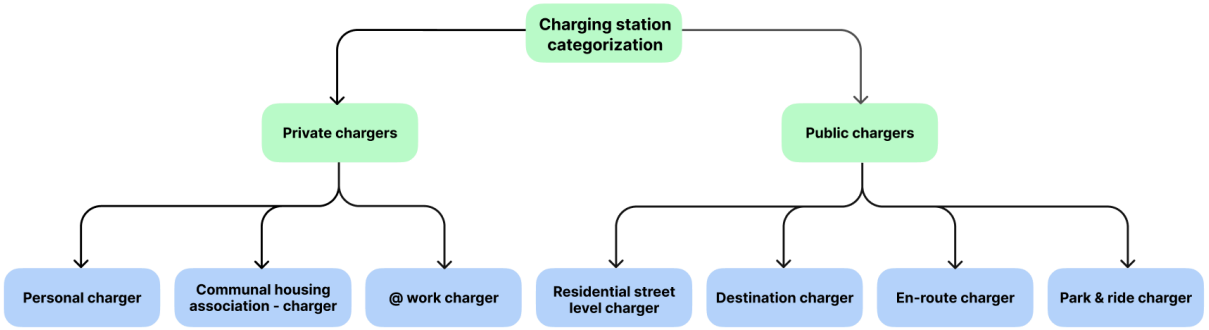


Figure 10. Overview of the different charging categorizations.

The different types of chargers are categorized by their use either at home, adjacent to a building for private use, or as public chargers available for everyone. These include:

Personal charger – a charger that the consumer owns and operates themselves.

BRF shared charger – one or more chargers that the housing cooperative (BRF) owns and operates, available for use by its members.

At-work chargers – one or more chargers that the organization owns or leases, available for use by its employees or individuals with access to the workplace.

Residential street-level chargers – chargers owned and operated by various public charging companies (CPO), which have agreements with the landowner or municipality to operate chargers at those locations.

Destination chargers – chargers located in close proximity to commercial buildings or businesses, such as shopping malls, amusement facilities, restaurants, and similar venues.

En route chargers – chargers located near highways, intersections, and major transport hubs. They are operated by a public charging company (CPO) and are mostly fast chargers or superfast chargers.

Park and ride chargers – chargers located in parking spots adjacent to public transportation stations. They are operated by a public charging company (CPO) and are used when traveling to and from public transportation stations.

One interesting takeaway from these observations was that, for both low-speed chargers and fast chargers across the inner city of Stockholm, there were many cars, both EV and internal combustion engine cars, parked at charging locations without being plugged in. This results in the mobile application, hereafter referred to as app, displaying the spot as "free" or "unoccupied," while in reality, the spot is occupied, as seen in Figure 11.

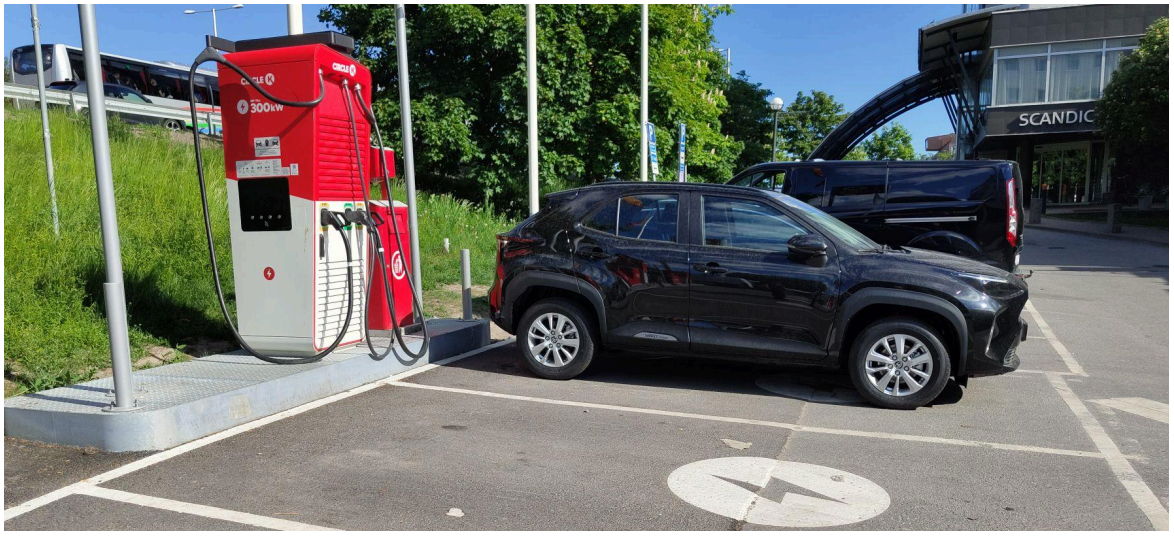


Figure 11. A car parked in the charging spot without being plugged into the charger at a superfast charger at an en route charging station (Circle K).

Another takeaway is the variety of interaction possibilities on different chargers. Some only have instructions on how to download the corresponding app and control the charger from there, while others have a screen with digital or physical buttons to control the charger. Most rely on a combination of the car's infotainment screen (or similar), the app, and an indicator on the charging station to communicate whether the charge has started, is ready to initiate, or is faulty, as shown in Figure 12.



Figure 12. A public residential street level charger with LED indicators.

An additional takeaway is that some chargers, primarily slow chargers, do not have cables installed (see Figure 12), while others do, requiring the user to carry an extra cable in the trunk of the car, just in case. Moreover, there are no indicators in the corresponding app to inform the user if they need to bring their own cable or not. In total, the observations took place at 15 locations across the greater Stockholm area, the different locations can be seen in Table 4.

4.2 User interviews

After the initial observations, user interviews were conducted. The interviews were semi-structured pop-ups held in the greater Stockholm area, at various types of public chargers, where users were actively charging their cars. By interviewing users during the act of charging, valuable insights and observations were gathered. A total of 25 individuals were

interviewed, including users charging their personal cars and those charging commercial vehicles as part of their work. Each interviewee's car model, type of charger, preferred interaction point, behavior, and experience were documented. The locations visited, types of chargers, and the number of interviewees are shown in Table 4.

Table 4. Locations visited, type of charger and the number of interviews conducted.

Location:	**Type of charger:**	**Interviews:**
Arninge	Destination charger + En route charger	0
Circle K Galoppfältet	En route charger	6
Circle K Solna Järva krog	En route charger	4
Gustav III:s Boulevard	Residential street charger	0
Kungsträdgården	-	2
Mall of scandinavia	Destination charger	1
Max Norrtälje	Destination charger + En route charger	6
Mcdonalds Hägernäs	Destination charger + En route charger	0
Narvavägen	Residential street charger + En route charger	2
OKQ8 Roslagsstoppet	En route charger	0
Quality Hotel Strawberry arena	Destination charger	0
Riddargatan	Residential street charger	0
Scandic Järva Krog	Destination charger	0
Strandvägen	Residential street charger + En route charger	2
Täby Centrum parking	Destination charger	2

The locations visited varied in category as well as in geographic setting. The interviews were conducted at different times of the day, including morning, after rush hour, midday, afternoon, and evening, on weekdays, weekends, and a holiday (Ascension Day, in swedish:

“Kristi Himmelsfärdsdag”). To support the interviews, a questionnaire was used as a foundation to spark discussion and facilitate shared insights. The questionnaire can be found in Appendix B. The results of the interviews were clustered and analyzed.

4.3 Interviews with EV charger retailers

To better understand the perspective of an EV charger provider, an interview was conducted with a retailer of EV charging stations, solutions, and services at *Eways* in Täby Centrum. The interview lasted about 30 minutes and provided valuable insights into the reasoning and behavior of EV charger buyers from a seller's viewpoint. One notable takeaway from this interview was the seller's experience with customers' most common concerns when considering going electric. These concerns included range anxiety, fear of occupied public chargers, and insufficient fast chargers along destination routes. Another interesting topic was the positive outlook on future developments in the expansion of the public charging network, he noted that many new charging stations are being brought online each week.

4.4 Interviews with EV retailers

Interviews with EV car retailers were conducted at *Hedin Bil*, where the car manufacturer *KIA*’s new EVs were showcased, as well as at *NIO*, a Chinese car manufacturer with a different approach to EVs, at Täby Centrum. *NIO*’s approach to EVs is that, instead of only charging the battery, the user can replace the entire battery pack while traveling; in this way, the user would receive a fully charged battery in a few minutes. This technology is relatively new and unique in the Swedish market, and *NIO* currently has only two stations where a battery swap is possible. The idea is to offer a complement to the public charging network, enabling users to swap batteries as needed on longer trips. The interviews aimed to gain insight into users' concerns, pain points, and challenges. The results of the interviews were used in the analysis.

4.5 Market analysis

A comprehensive market analysis was conducted on the most common mobile applications used for public EV charging. Different functions were analyzed and mapped, including the registration procedure, integration with home chargers, and payment methods. The various payment options were categorized into two sections: pay at the charging point, physically, or

pay via a digital solution, either through an app or via the car's infotainment system. The different payment options are illustrated in Figure 13.

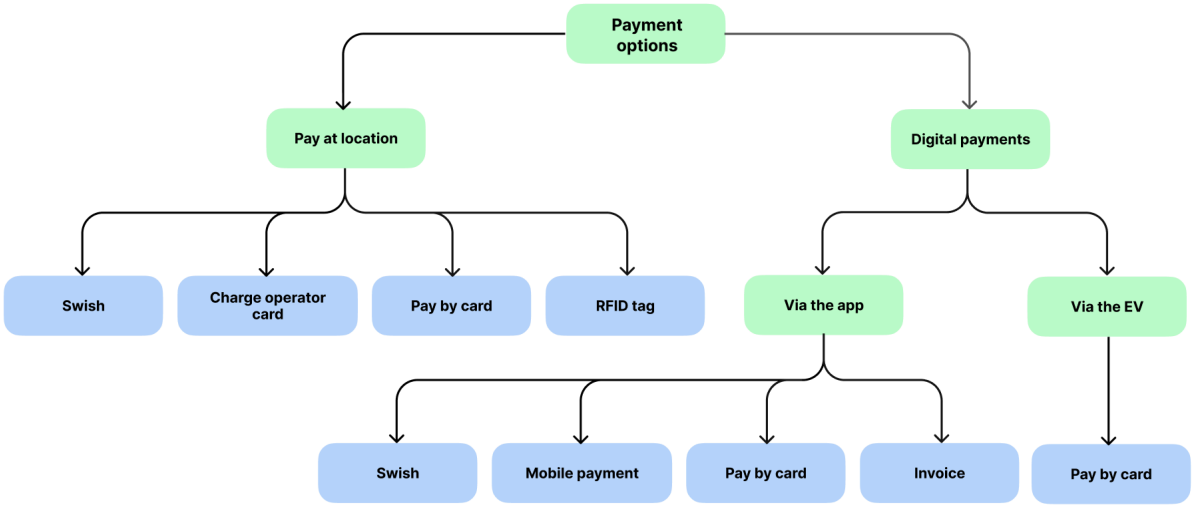


Figure 13. Categorization of the various payment options.

At the charging point, when paying at physical locations, there are up to four different payment options: Swish, a credit/debit card, an RFID tag, or a specific charging card issued for one or more charging stations. The charging card may offer various payment options. When paying digitally using an app, there are also up to four different options: Swish, mobile payment (e.g., Apple Pay/Google Pay/Samsung Pay), invoice, or credit/debit card. When paying digitally via the car's system, a credit/debit card can be used and must be set up with the car in advance.

The most common apps analyzed included Fortum Charge & Drive, Incharge, Qwello, Mer Sverige, Circle K Charge, Ionity, Northe, Monta, OKQ8, and Elli, covering both CPOs and eMSPs. The apps were analyzed in depth individually, with functions, pros and cons, layout, etc., listed for each. Table 5 presents a general preview of the analyzed apps in comparison with each other. A blurred overview of the full comparison is available in Appendix C.

Table 5. Analyzed apps and a sample of key differences regarding plug & charge, payment methods, account requirements, partnerships, and integration with home chargers.

Function: / APP:	Plug & Charge	Payment methods	App use without an account	Partnerships	Integration with home chargers
Fortum Charge & Drive	No	Card, Mobile payment	No	Yes	No
Incharge	No	Card	Yes, but limited	Yes	No
Qwello	No	Card, Paypal	Yes, but limited	No	No
Mer Sverige	Not at the moment.	Card	Yes	Yes	No
Circle K Charge	Yes	Card, RFID, Mobile payment	No	Yes	No
Ionity	Yes	Card	No	No	No
Northe	No	Card, Mobile payment	No (Bank-Id available)	Yes	No
Monta	Yes	Card, RFID	No	Yes	Yes
OKQ8	No	Card, RFID	Yes	Yes	Yes
Elli	Yes	Card, RFID	Yes	Yes	Yes

Partnerships refer to agreements made by the app owner (eMSP) with other CPOs to display their charging stations in the eMSP's app.

Notably, the payment methods across different apps vary greatly, and some apps linked to specific chargers also offer payment via RFID tags. Another insight is that only a handful of apps include the option to connect to home chargers. As a result, from an user perspective, the multiple functions can feel overwhelming and complicated. Additionally, some apps lack partnerships, displaying only their own chargers, while the same company may have partnerships with other CPOs and display their chargers in those apps.

The functional differences analyzed were as follows:

- Registration
- Payment Options
- Filters and Search Functions
- Plug & Charge
- Integration with home chargers
- Car information
- Travel planning
- Services for BRFs and companies
- App use without an account registration
- Partnerships
- Reservation of chargers
- Subscriptions / Loyalty program
- Analyzed user data

The results of the market analysis were incorporated into the analysis and connected to the user analysis.

4.6 Stakeholder map

The stakeholders of this project are those with an interest in the project's outcome. The project focuses on the greater Stockholm region, although the findings may also be applicable elsewhere. The stakeholders are based on the actors in the network described in chapter 2.1 EV charging technology, as well as additional stakeholders, all mapped in Figure 14, with a description in Table 6, indicating the level of interest and influence each stakeholder has.

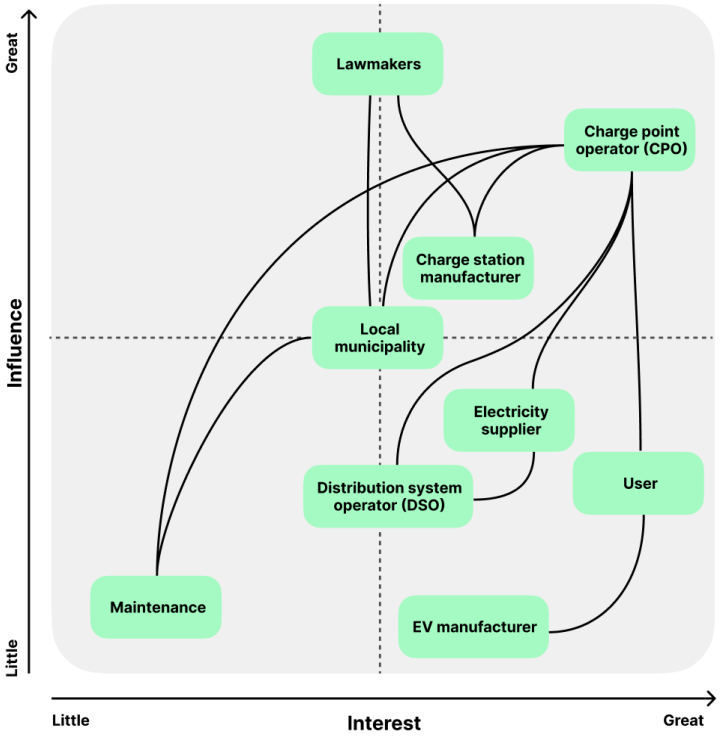


Figure 14. Stakeholders of the project and the amount of interest and influence they have.

Table 6. The stakeholders: a description, their interest, and influence in the project.

Stakeholder:	Description:	Interest and influence:
Lawmakers	Both government and EU lawmakers.	Have great influence on the project and their interest is mildly high.
Charge Point operator (CPO)	The charge point operator, in this case VCG.	Have great influence on the project and also great interest.
Charge station manufacturer	The manufacturer of chargepoint stations.	Have a mildly high influence on the project and their interest is mildly high.
Local municipality	In this case Stockholm city.	Have somewhat high influence on the project and their interest is somewhat high.
Electricity supplier	The retailer of the electricity used.	Have light influence on the project and their interest is mildly high.
Distribution system operator (DSO)	The grid operator. The company that owns and maintains the grid.	Have light influence on the project and their interest is somewhat high.
User	The customer using the charge point.	Have light influence on the project and their interest is great.
Car manufacturer	The manufacturers that produce EVs.	Have little influence on the project and their interest is mildly high.
Maintenance	The entity doing maintenance on the physical charge station.	Have little influence on the project and their interest is little.

The *lawmakers* have great influence on the project, as they are responsible for deciding on legislation and laws, and their interest is mildly high. They are connected to the *local municipality* and *charge station manufacturer*, as these entities primarily follow and implement said legislation and laws. The local municipality and charge station manufacturer are connected to the *charge point operator (CPO)*, in this case, VCG, which has both great influence on the project and also great interest. The local municipality is also connected to the entity responsible for *maintenance*, as they need to collaborate. The entity responsible for

maintenance is, in turn, connected to the CPO. The CPO has a strong connection to the users, who are also affected by the *EV manufacturer*. Additionally, the CPO has a strong connection to the *electricity supplier* and *Distribution System Operator (DSO)*, as all three must collaborate to deliver energy to the user.

5. Analyzing the field data

This chapter focuses on presenting the analysis of the user studies and market analysis. Different methods were used to analyze the data and generate insights, including the creation of need groups and customer journey mapping (CJM), after which insights were developed. These insights provide a foundational understanding of the user, market, and technology and will be translated into solutions in the design phase of the project.

5.1 Clustering

After the interviews with various users, retailers, and others, the responses were clustered using post-its on a large digital board in *Figma*. The post-its were then sorted into sections based on the type of quote or response. Clustering the different responses from the interviews led to the discovery of multiple insights, including both general insights and more specific insights, as well as tensions.

From the clustering, four interesting themes of feelings were identified. The themes include quotes translated from Swedish to English. The original quotes can be found in Appendix D.

The four themes identified were:

A feeling of "**uncertainty**"

> *"It's usually occupied, and you need to account for the distance if you're driving with not that many kilometers left."* - Respondent P17

Users have experienced a lack of trust in the information provided about charge point availability or whether the charge point is operational. The information they receive via the app or the car is not always accurate. As a result, users feel doubt regarding public EV charging; they cannot trust the infrastructure and therefore experience a feeling of uncertainty.

A feeling of "**powerlessness**"

> *"Had to call support because I couldn't stop the charging via the app, and couldn't remove the charger."* - Respondent P15

Users have expressed difficulties in managing a charging session, with problems such as being unable to start a charging session or disconnect the car from the charge point. This has led users to feel trapped or deceived, resulting in a feeling of powerlessness.

A feeling of "**confusion**"

> *"[...] sometimes I feel that there are too many features in the apps, so you almost become confused."* - Respondent P3

Users have expressed how cumbersome some apps are to navigate; they have to scroll through multiple interfaces or switch between different apps just to start a charging session. This has led users to feel frustration regarding certain aspects of the public charging experience, resulting in a feeling of confusion.

A feeling of "**disorder**"

> *"[...] however, there are apps that don't launch BankID directly, which means you need to exit the app, find BankID, identify yourself, and then return to the app."* - Respondent P14

Users have expressed how turbulent the public charging experience is, citing the need to navigate multiple different apps. The apps themselves lack understandability, requiring users to switch between apps manually just to proceed with an action. This has caused users to feel instability within the public charging space, resulting in a feeling of disorder.

The clustering process revealed some interesting tensions, with users expressing differing viewpoints on certain issues.

The identified tensions are:

understandable payment options	———	frustrating payment options
always available	———	always occupied chargers
cheap	———	expensive
mobile applications always work well	———	mobile applications never work

The tensions are closely linked to the type of user responding; more experienced users perceive solutions as less challenging, while less experienced users believe the opposite.

5.2 Need groups

The need groups are based on the results of the interviews and the general insights from the clustering. The segmentation was iterated back and forth, and the final version of the customer segmentation is based on needs and behavior. Figure 15 displays the need groups on two axes: commercial-individual and dependent-flexible.

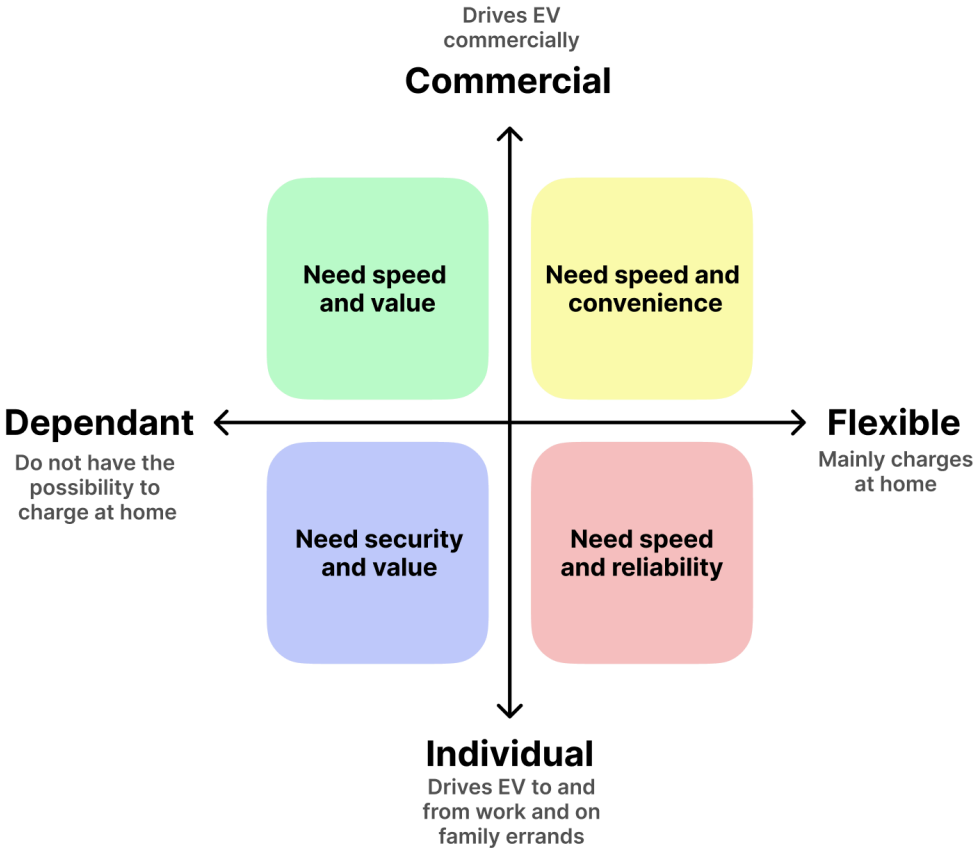


Figure 15. Segmentation of users based on needs and behavior.

Users who drive *commercially* and are *flexible* need speed and convenience. This means that, since they drive commercially, they need fast chargers to quickly recharge and continue, while also requiring convenient charging locations. As they drive frequently, they aren't particularly loyal to any specific location but rather look for the fastest charger available in the area. They are not highly price-sensitive when charging publicly, as they primarily charge at home or can bill their employer.

Users who drive *individually* and are *flexible* need speed and reliability. This means they are not particularly price-sensitive when charging publicly, as they primarily charge at home, and therefore choose whichever fast charger is nearby or along their route. They are demanding users who expect chargers to always work and are quickly frustrated if they do not. Additionally, they need chargers to be fast, as they are accustomed to charging at home and tend to be impatient.

Users who drive *individually* and are *dependent* need security and value. This means they are price-sensitive, as they do not have access to a charger at home. They are therefore vigilant, seeking a secure, reliable charger with good pricing, located either close to home, along their route, or near their workplace. Once they find a charger they like, they remain loyal to it, incorporating it into their daily routine for regular charging. Thus, they need the security of having a reliable, accessible charger.

Users who drive *commercially* and are *dependent* need speed and value. Like commercially flexible users, these users also drive commercially and require fast chargers to quickly recharge and continue, while also needing convenient charging locations. As they drive frequently, they aren't particularly loyal to any single location but instead look for the fastest charger in the area. At the same time, they seek the lowest-priced charger, as they lack access to a charger at home. Thus, they are vigilant, aiming to find the best of both worlds: speed and value. They have tried various charging operators, but once they find one or more operators with good locations, fast chargers, and relatively low prices, they tend to stick to those operators.

5.3 Customer journey mapping

A CJM (Customer Journey Map) was created based on the identified need groups to further explore touchpoints, needs, pains, feelings, and opportunities. The different need groups were listed on one axis, and the different "Journey Steps" (i.e., the steps the user encounters during the experience) on the other axis. The journey mapped represents a "normal" charging experience, beginning when the user notices their battery is running low and ending when they leave the charging location ("normal" in the sense that the charging session takes place at a standard charging station with multiple charge points). Additionally, "Actions" were also mapped, addressing questions like: What does the customer do? What information do they seek? What is their context? For example, the action could be "Navigates to charge location".

One interesting takeaway from the CJM is the pain point associated with trying to start a charging session; having to interact with the app to initiate a session is cumbersome. This applies to all need groups, as this interaction “action” is inconvenient regardless of the user. The complete CJM can be found in Appendix E.

5.4 Brand analysis

The brand analysis was initiated after finding that the proposed service extension closely aligns with the company's mission, vision, offerings, and, by extension, the company's brand. Therefore, a brand analysis was conducted by interpreting eight different aspects of a brand expressed in a brand identity. Figure 16 shows the eight different aspects of a brand identity.

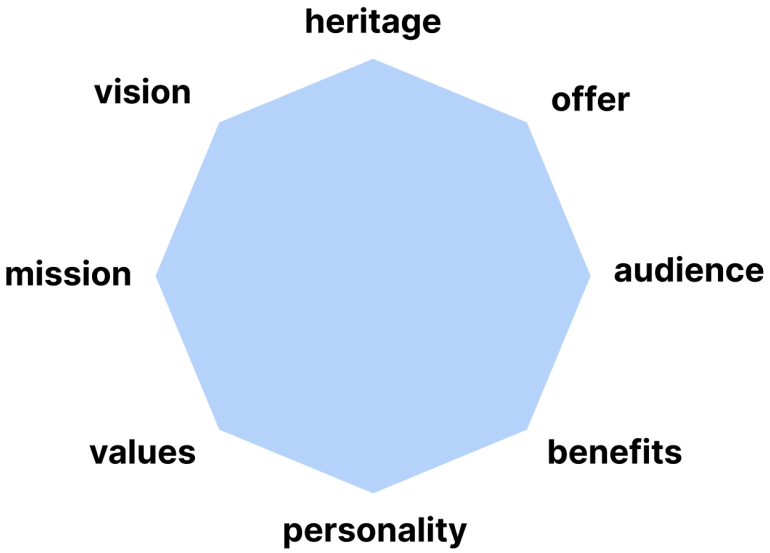


Figure 16. Eight different aspects to a brand identity.

The analysis of the brand identity of the mother brand laid the foundation for the brand platform as a stepping stone towards a new brand extension.

The results of the brand analysis have been removed from the report due to confidentiality.

5.5 Insights

The final step in analyzing the field data was to formulate insights. The insights are based on the interviews, market analysis, clustering, need groups, CJM, and brand analysis. The purpose of the insights is to compile information for the continuation of the design phase, as well as to highlight the main conclusions of the study to be handed over to VCG. The insights

consist of seven major insights, with supporting smaller insights included within them. Translated quotes from Swedish to English are included, with the original quotes available in Appendix D.

Insight 1: The Blind Spots of Expertise: Experienced users are unaware of today's challenges.

While different users experience the challenges of today's status quo differently, the most experienced users seem to have overlooked the difficulties of current solutions. They are accustomed to a complex solution and do not encounter the same challenges as less experienced users.

> *"It's okay, I can charge everywhere"* - Respondent P7

Said one interviewee as he proceeded to show the multiple RFID tags and cards connected to different charging operators, all stored in a box in the center console of his car.

> *"It works fine, I have all the things I need"*- Respondent P5

One interviewee remarked, then proceeded to show the folders on his phone containing multiple apps for different charging locations and apps for finding chargers. He opens several apps to complete a single action, accustomed to complex solutions.

Having used the public charging network for quite some time, these users have had time to adapt to the solutions at hand.

Insight 2: Saturated mobile application market leads to disorientation.

This insight is based on the fact that the mobile application market for public charging is highly saturated. There are many different operators, most of which have their own proprietary mobile applications that service their own chargers, while supporting a few other charge operators as well. This results in users needing to have many different apps from different operators in order to charge at all the charging stations.

> *"There are way too many apps. Some don't work, some work better. Regardless, it's a messy, hard-to-navigate jungle."* - Respondent P3

With an overwhelming number of apps, each offering varying levels of functionality and compatibility, EV users often face confusion, especially among inexperienced users. This abundance of platforms makes it difficult to find reliable charging stations, navigate different payment systems, and seamlessly integrate with charging networks. As a result, what should be a straightforward process becomes a frustrating, time-consuming task, hindering the adoption of electric vehicles. There is a clear need for greater clarity in the market, as one respondent said.

> *"I would like to have an app for all [public charge points], maybe there is but I don't know about it anyway"* - Respondent P9

Insight 3: Protocols between the EV, charging station, and operator are not always interconnected.

Calling support when dealing with public charging appears to be an inevitable part of the experience, especially for older or newly released cars.

> *"It feels like it never works. I have to call support three out of every fourth time [when using public EV charging] "* - Respondent P9

The problem seems to lie in the communication protocols between either the charge point, the EV, or the operator.

> *"I had a KIA before and it always had problems disconnecting the cable itself from the car after I had fully charged it. Had to call the operator and ask them to disconnect me all the time."* - Respondent P1

The problem, when it occurs, results in great frustration and a feeling of powerlessness. This issue of being unable to disconnect was experienced by quite a few of the respondents, but not all. On the other hand, they also had to contact support at one time or another for other reasons; for example, the charge was unable to start, or the charger did not recognize the car.

Insight 4: Varying parking rules across municipalities impact assumptions about charging point availability.

Differences in parking regulations across municipalities in Sweden create uncertainty for EV users regarding charging point availability. In some municipalities, parking at a charging

station is restricted to EVs actively charging, while in others, non-charging vehicles are allowed to occupy the space. These inconsistencies lead users to assume that charging points may not always be accessible, even when they appear available, affecting their charging decisions and travel planning.

> *" [...] you can't trust that there is always room when you arrive, or that they work as they should for that matter."* - Respondent P11

This leads to users anticipating that it won't always work out as planned, resulting in the need to plan for some level of uncertainty.

> *"It's usually occupied, and you need to account for the distance if you're driving with not that many kilometers left."* - Respondent P17

As a consequence, users factor in potential detours or delays to ensure their charging needs are met, making the experience more stressful and less predictable.

Insight 5: Range anxiety is amplified by a lack of trust in the infrastructure.

This insight is linked to the previous one regarding how varying parking rules across municipalities impact assumptions about charging point availability, creating uncertainty and, in turn, range anxiety.

Range anxiety, a common concern among EV users, is significantly amplified by a lack of trust in the charging infrastructure. Many users express doubts about the reliability, availability, and functionality of public charging stations. Factors such as inconsistent maintenance, unexpected outages, or inaccurate station information in apps lead users to fear that they may not be able to charge when needed. For example, one interviewee experienced misinformation about the need to bring one's own cable to the charge point.

> *"Shit, there wasn't a cable here, I'll check if I have one, [starts looking in the trunk for a charging cable] [realizes she doesn't have the right cable with her], I'll have to drive on"* - Respondent P12

This lack of trust in the infrastructure leads to heightened anxiety about running out of battery, prompting users to over-plan their trips or avoid longer journeys altogether, even when their vehicle's range could otherwise accommodate such trips.

Insight 6: Price of public EV charging is a driving factor in selecting charging locations.

The price of public EV charging plays a critical role in how users select charging locations. Some EV owners are highly cost-sensitive, often prioritizing affordable charging locations over convenience or proximity. This sensitivity to price leads users to compare charging rates across different providers, sometimes opting for slower chargers or less convenient locations if they offer a better rate. This is especially true for users who do not have access to a home charger, as one commercial driver dependent on public charging said:

> *"I always try to find the cheapest charger nearby"* - Respondent P11

As a result, pricing transparency and competitiveness have a direct impact on user decisions, with high costs potentially deterring the use of certain charging stations and influencing travel behavior. The market analysis also highlighted the lack of price transparency in some of the apps, underscoring the problem.

Insight 7: The user experience in some mobile applications linked to public EV charging is of poor quality.

The user experience in some mobile applications is poor, causing frustration for EV owners. Issues such as confusing interfaces, slow response times, and inaccurate data about charging station availability or prices hinder the charging process. Many apps lack intuitive navigation, making it difficult for users to find essential information like charger types, real-time status, or payment options. Additionally, some apps include too many niche features.

> *"[...] sometimes I feel that there are too many features in the apps, so you almost become confused."* - Respondent P3

Furthermore, unreliable app performance, such as frequent crashes or incorrect station details, can lead to wasted time and inconvenience.

> *"[...] however, there are apps that don't launch BankID directly, which means you need to exit the app, find BankID, identify yourself, and then return to the app."* - Respondent P14

These frustrations detract from the overall charging experience, leaving users dissatisfied and more unlikely to trust or rely on that particular app for their charging needs.

6. Design phase

This chapter explains the concept development phase, including ideation and prototyping. In addition to the concept, a brand extension was formulated alongside the final concept.

6.1 Target group

The design specifications are a result of the insights and the targeted user need group. Among the defined need groups, one stood out as the most relevant. The chosen need group to focus on was those who need *security* and *value*. The decision to focus on this need group was based on the fact that it is underrepresented in the field and presents an interesting opportunity. This group also experiences most of the challenges with today's solutions, as they are highly dependent on it working effectively, as discovered in the CJM.

6.2 Ideation

The ideation phase mainly consisted of a brainstorming activity within the team. The process began by individually writing down and sketching different ideas, before presenting them to one another and discussing each one. Following this, a second ideation session was conducted based on the initial ideas. The ideation phase resulted in a variety of ideas to address different aspects of the problem. *Figma* was used to sort and cluster the ideas; see Appendix F for a full view of the different ideas plotted on a whiteboard. These ideas laid the foundation for the ongoing design process, leading into rapid prototyping of various concepts.

6.3 Brand extension

As part of the design development, and in conjunction with the design of the mobile application, a need for a brand extension was identified. The purpose of the brand extension is to align with the outlined service, making the mother brand's value proposition more coherent and providing a natural step toward implementing the service. The brand extension is based on the brand analysis defined in Chapter 5.4, Brand Analysis. This extension incorporates three categories, as seen in Figure 17—mission, offer, and audience—from the identified analysis into the final concept.

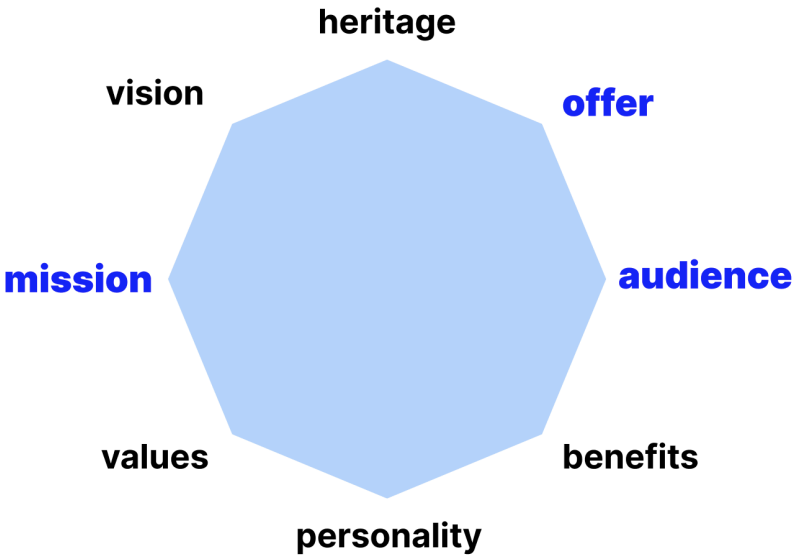


Figure 17. The three identified aspects that need transitions of a brand identity.

The results of the brand extension have been removed from the report due to confidentiality.

6.4 Rapid prototyping

To test the different ideas quickly from the ideation stage, rapid prototypes were created using various materials such as paper, card stock, pens, and scissors. The goal was to experiment with different layouts, experiences, solutions, and outlines. The rapid prototypes produced layouts that could be evaluated quickly and allowed for the optimization of characteristics such as shape, size, and overall usability. Figure 18 shows three examples of layouts resulting from rapid prototyping.

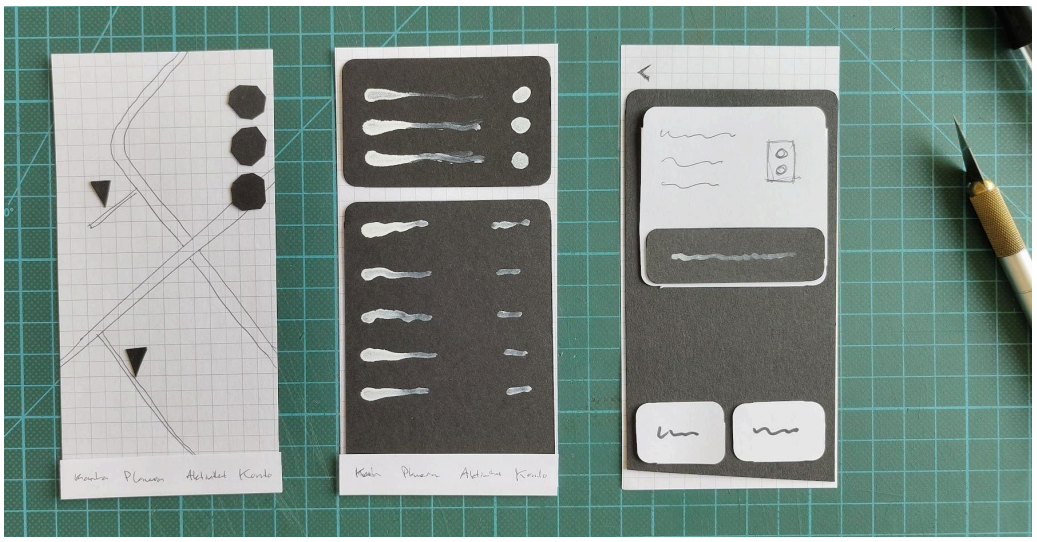


Figure 18. Examples of layouts of rapid prototyping.

The outcome of the physical prototyping led to the use of frames laid on top of one another. As the user navigates through the app, frames appear with additional information and actions. The results of the rapid prototypes were then translated into digital tools and further iterated before the final design of the mockup was completed.

6.5 Concept development

The concept development phase was carried out in *Figma*, where ideas from the ideation phase were conceptualized into various concepts and solutions. The concept development resulted in the formulation of one main concept with key functions highlighted. The functions were defined based on the analysis and organized into five solution topics derived from the insights about the identified problems. The overall functions, opportunities, and solutions are presented under each solution topic:

Transparency

To give users more control and insight into the charging systems, increased transparency is needed. Transparency regarding the operator, pricing, support access, availability, and cable indicators is crucial. These solutions are linked to insights 2, 5, 6, and 7. Additionally, it addresses feelings of “uncertainty” by providing users with more up-to-date information about availability and charge point operability, as discussed in Chapter 5.1.

Ease of use

To assist users in utilizing the public charging infrastructure, there is a need for easy-to-use applications. That is, having a smooth registration process, being easy to understand, offering enough payment options, and assisting users in finding the right charger with useful filter functions. Additionally, there should be a function to help users plan their trips effectively. The concept should include technology to register a user's EV, which will help plan effective trips based on the performance of their EV. These solutions are linked to insights 1, 2, 6, and 7. In addition, it addresses the feeling of “disorder” by streamlining the process and eliminating unnecessary steps.

Implement smart charging

Through the implementation of smart charging, users can gain a sense of control over the charging process in terms of both cost-effectiveness and charging requirements. Additionally, users are able to lower their cost per charging session while also satisfying the power system.

Both the user and charging operator can benefit from smart charging technologies such as peak shaving and energy arbitrage. Peak shaving reduces energy drawn from the grid during times of high demand by adjusting EV charging schedules or implementing mathematical algorithms. This approach helps prevent grid overload and, most importantly, lowers energy costs for users charging their vehicles, which is especially beneficial for the chosen target group, as discussed in Chapter 2.2 Smart Charging.

The implementation of smart charging is enabled by communication protocols like ISO 15118 and OCPP, as discussed in Chapter 2.3 Communication protocols. The control modes supported in ISO 15118-20 enable smart charging, increasing the user's sense of control by offering modes where drivers can either directly manage the charging process, considering only grid restrictions, or communicate their charging needs to the system, allowing the CPO or eMSP to create a suitable charging schedule based on those requirements. OCPP also plays a key role in implementing smart charging by facilitating communication with the charging station management system. This requires the latest versions of ISO 15118 and OCPP, specifically ISO 15118-20 and OCPP 2.1 as of this date.

This solution is linked to insights 3 and 6. Additionally, it addresses the feeling of “powerlessness” by giving users more options regarding the price of charging their EV.

Implement plug and charge

By implementing the Plug & Charge function, we can eliminate one of the most frustrating steps: “standing in front of a charger and trying to start/stop a charge via the app.” Although not all chargers or EVs support this function currently, the benefits are so profound that it should be implemented. The Plug & Charge feature is enabled by the ISO 15118 standard. This solution is linked to insights 3 and 7. Additionally, it addresses feelings of “confusion” by eliminating the need to control charging via the app and “powerlessness” by reducing friction in managing a charging session.

Charging locations

There is a need to assist users in navigating charging locations, giving them more control and insight, especially for the chosen target group that seeks value and security. Features to assist the user include the ability to create favorite charge locations, view forecasts for those locations, access a charging status widget, and, most importantly, have all charging stations consolidated in one place. This will reduce the number of applications needed to navigate the public charging space. These solutions are linked to insights 1, 2, and 5. Additionally, the solution addresses the feeling of “uncertainty”, by giving users reassurance and more control over their charging habits.

The app focuses on the needs of the identified need group while also accommodating other identified needs. The overall layout of the mockup was developed using basic frames with key functions on each. An overview of the proposed layout of the mockup can be seen in Appendix G.

7. Final concept

This chapter presents the results of the final design, detailing the outcomes and solutions developed throughout the project. It explores how the design addresses the identified needs, challenges, and objectives, showcasing key features.

7.1 Final design

The final design of the mockup is based on the conclusions from the concept development phase. The mockup is constructed in a *Figma design file*, with individual frames showcasing the different functions deemed necessary. The overall layout is inspired by the "Charge and Drive" application, as well as the stacked frame approach developed during rapid prototyping. The design frames are based on an Apple iPhone 15 Pro, and therefore the aspect ratio follows that device's specifications. The language in the mobile application, and consequently the names of the different functions, is in Swedish.

The first frame when entering the app for the first time is the registration frame, where users can register by validating their BankID and adding their email, registering via their Google or Apple account, or by registering manually. Figure 19 shows the registration frame.



Figure 19. Frame of registration (first time only).

Once the user has registered, they are brought to the main frame in the tab called *Ladda*, which acts as a home screen, as shown in Figure 20.

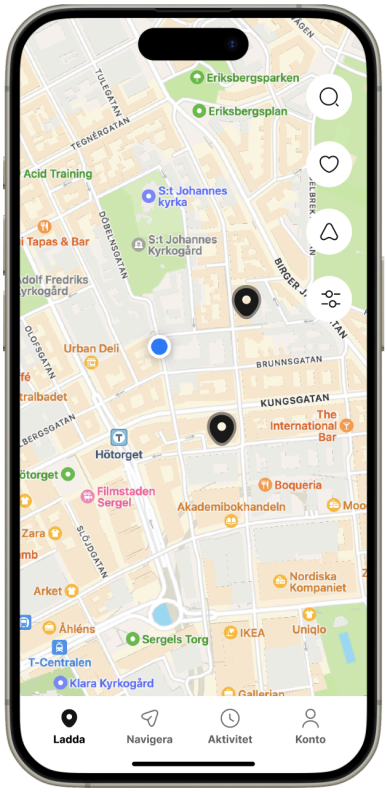


Figure 20. The main frame, *Ladda*.

Here, the user can view and navigate the map and all the different charging locations. There is a navigation bar at the bottom to access the four main tabs: *Ladda*, *Navigera*, *Aktivitet*, and *Konto*. The *Ladda* tab acts as a home screen. The interface includes a search function, a function for favorite charging locations, a function to reset the map to the user's location, and a filter function. Figure 21 shows the filters available under the filter function.



Figure 21. The different filters available to apply to the charge location map.

Here, the user can apply filters for charger speed, means to start a charge, preferred connectors, preferred charge operator, smart charging capability, and availability, as well as filter by price.

From the *Ladda* frame, the user can tap on a charge location to learn more about it. Once a charge location is tapped, a frame appears, as shown in Figure 22.

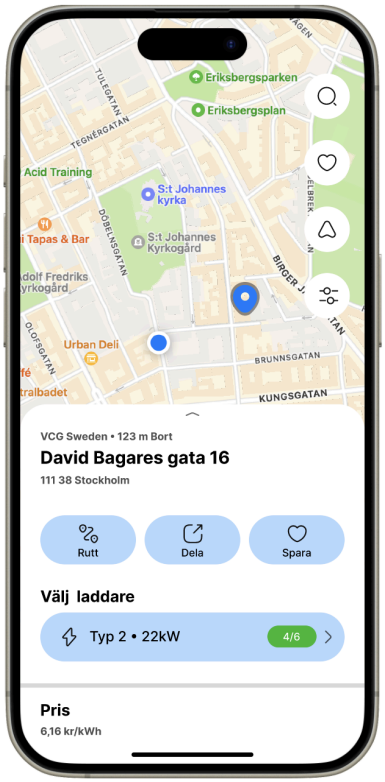


Figure 22. Information about a charge point displayed in a swipeable frame.

The information frame overlays the map on the *Ladda* screen and can be swiped up for more details. Here, the charging station operator, street address, and distance from the user's location are displayed. There is a function to navigate to the selected charge point, which opens the default map app installed on the phone (e.g., Google Maps, Apple Maps, or Waze). There are also share and like buttons; if the charger is liked, it appears under the user's favorites in the *Aktivitet* tab. The next section displays the type of charger and its charging power, along with the number of chargers available. If selected, a new frame appears where the user can choose a charge point and start a charging session. Scrolling up further on the information frame reveals additional details about the charge point, as shown in Figure 23.

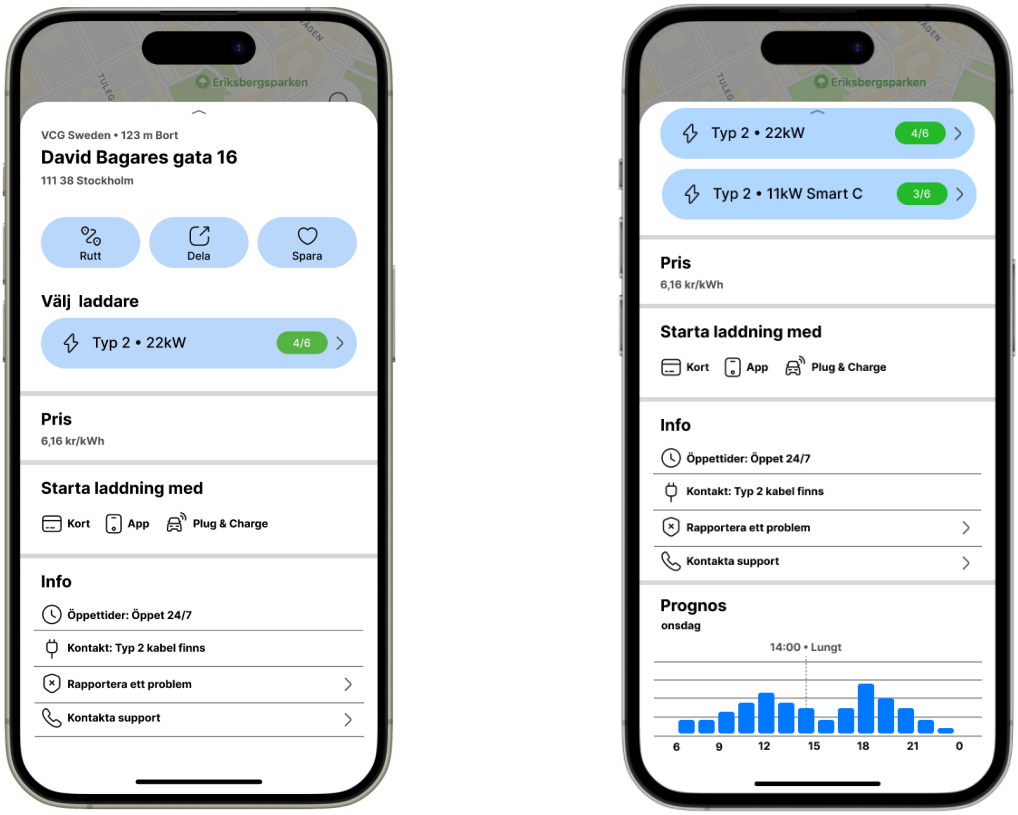


Figure 23. Additional information about a charge point. The left screen shows pricing, available means to start a charging session, while the right screen continues the scroll and displays availability forecasts.

Here, the user can view the price, expressed in kr/kWh. Following that, the user can see the available means to start the charging session: either by paying with a card at the charge point, starting and paying via the app, or using Plug & Charge. The next section displays information about the charge station, including opening hours, the type of connector required, whether a cable is needed, the ability to report a problem with the charger, and a function to contact support. The final section provides a forecast of charge point activity, showing peak occupancy times.

If a charging station supports smart charging, a second row of chargers will appear in the same section as the other charge points, as shown in Figure 24.

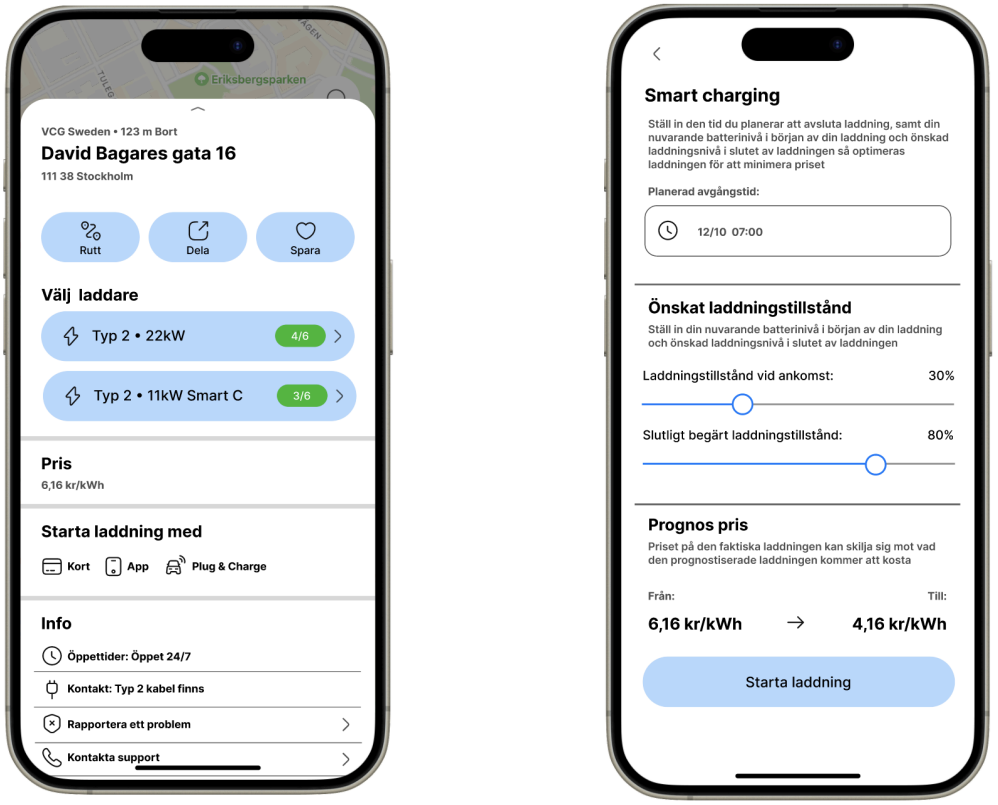


Figure 24. Smart charging frames. The left screen shows the available smart chargers, while the right screen shows the smart charging functions and the controls.

If a charger with smart charging capabilities is selected, a new frame labeled *Smart Charging* appears. Here, the user can set their estimated departure time, state of charge at arrival, and final requested state of charge. An estimate of the final price will also be displayed; the cost savings will vary depending on how much charge is needed, at which times, and what type of charger is used. Below that is a button to start the charging session.

Once a charging session has been initiated, a live lock screen notification will appear, as shown in Figure 25.



Figure 25. Live lock screen notification displaying the state of charge.

The notification displays the charging location, charger speed, charger type, the EV being charged (in this case, a Polestar 2), as well as the state of charge.

The second main tab, *Navigera*, is used for planning longer trips, where stops to charge the EV are planned for. If a user enters *Navigera* for the first time, a frame for registering their EV appears, as shown in Figure 26.

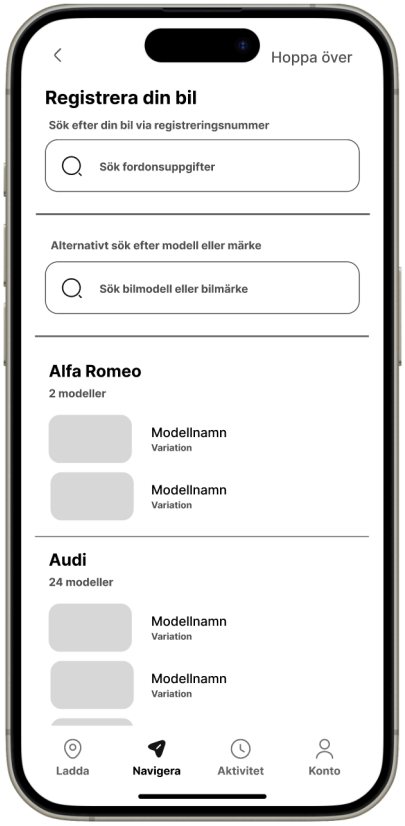


Figure 26. Frame for registering the user's car.

Here, the user can search for their car by entering the vehicle registration number, searching for the manufacturer and model, or manually scrolling through the list of EVs to register. The idea is that with the car's information gathered automatically, the route planning function can be much more accurate, taking into consideration the vehicle's performance, efficiency, battery size, charge connectors, and so on.

Once the car is registered, the route planning function appears, as shown in Figure 27.



Figure 27. The second main tab, *Navigera*.

Here, the user can enter their starting location, destination, and add potential stops along the route. Next, the user can enter the state of charge at the beginning of the trip as well as the preferred state of charge on arrival. Below that, the selected car is displayed. At the bottom is a button for planning the trip, which will compute the optimal route along with the necessary charging stops. The completed route will appear with information about the estimated travel time, number of stops required, charging locations, duration of each charging stop, and total energy consumption. Here, the user can click a button that will open the default map app installed on the phone (e.g., Google Maps, Apple Maps, Waze) with the route already planned.

The third main tab, *Aktivitet*, displays the user's favorite charging locations and history, as shown in Figure 28.

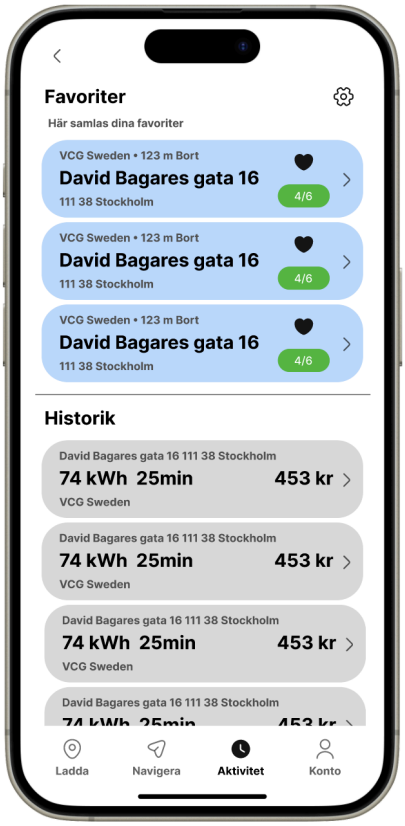


Figure 28. The third main tab, *Aktivitet*.

Here, the user can view their favorite charging stations. At a glance, they can see the different locations and their respective charge point occupancy. From here, they can tap on each location to enter an expanded view, as shown in Figure 23, with more information and access to the different charge points. In the next section, called *Historik*, the user's charging history is displayed. Each charge session is shown with details about the charge location, energy consumed, duration, and final price. From here, the user can tap on each session to view their receipt.

The fourth main tab, *Konto*, contains the user's profile, information, and settings, as shown in Figure 29.



Figure 29. On the left: the fourth main tab, *Konto*; on the right: a frame of the payment options.

Here, the user can access their account to change payment options, e.g., add a card, Apple Pay, Klarna, or Swish, and view their personal information. They can also adjust settings related to their car, Plug & Charge, Smart Charging, and overall app and account settings. Additionally, users can access in-app support and find detailed information about the app if needed.

The overall functions and features implemented in the mobile application mockup address the insights and problems identified in Chapter 6.5 Concept Development.

7.3 New brand extension

As part of the results, a new brand extension is presented. Visualized on the findings from the brand analysis.

The results of the new brand extension have been removed from the report due to confidentiality.

8. Discussion, conclusion and recommendations

This chapter presents the discussion, conclusion, and recommendations derived from our findings. The discussion highlights key insights and addresses any limitations encountered during the study. Following this, the conclusion synthesizes the overall findings. Finally, recommendations and future work are presented based on the analysis, offering practical next steps to advance the field and inform future work.

8.1 Discussion

The project concluded with the development of a mobile application mockup that presents functionalities designed to meet the identified user needs, as determined in the study, and is supported by technologies highlighted in the literature review.

Overall, the solutions presented in the mockup strike a balance between multiple functionalities and the ease-of-use of a mobile application. On one hand there is a need for multiple functionalities, and on the other hand there is a need for intuitive, ease-of-us mobile applications without it being too clustered. On one hand, there is a need for multiple functionalities, and on the other hand, there is a need for an intuitive and easy-to-use mobile application that is not too cluttered. This challenge of balancing user needs with the overall usability of the app presented a notable tension. In the end, the proposed design achieves this balance, by using the four main tabs to guide the user, along with overlaid frames appearing on top of the active frame, which reduces clutter. This ensures that the functions a user is searching for are quite intuitive. Moreover, the careful use of frames efficiently guides users through different stages of the app, enabling them to complete specific tasks more effortlessly.

Another challenge was that some users viewed the ability to avoid logging into the app as a positive feature. However, in our proposed mockup, logging in is required. The idea here is that, since the mockup includes several functions requiring users to input their personal details, a single global login upon the user's first entry to the app strikes a good balance. This approach ensures that users only need to log in once, or when they actively log out, enabling seamless access to all the app's functionalities.

Among the defined need groups, one stood out as the most relevant: those who need security and value. The decision to focus on this need group was partly based on the fact that they are

underrepresented in the field, face most of the challenges with today's solutions, as identified in the CJM, and because this group presents an interesting opportunity. As EV batteries continue to decrease in cost, the vehicles themselves have also become more affordable in recent years and are expected to become even cheaper in the future. Consequently, the typical EV driver will likely be younger and less affluent than today's drivers. This trend suggests that this need group will grow significantly in the future, making it even more compelling to address their needs.

Another important consideration is that the basic needs of this chosen group align well with those of other need groups. However, their more specific needs for better value are addressed by offering additional charging options, enabled by smart charging capabilities through communication protocols. These protocols allow for dynamic pricing and load management, improving the overall user experience but providing particular benefits to users who rely on public charging and are price-sensitive.

The communication protocols also introduce features such as plug and charge, which streamline the charging process and enhance user satisfaction. One of the major hurdles users face when dealing with public charging, as revealed in the CJM, is the process of plugging in their car and attempting to start or stop a charging session via the mobile application. By introducing plug and charge, this step can be eliminated entirely. On the other hand, if a user opts for smart charging, they will still need to control the charging process through the mobile application.

On a related note, there is a pressing need for more standardized practices within the industry to streamline the user experience and ensure greater uniformity in both charging stations and digital interactions. While some might argue that increased standardization could stifle innovation, well-designed practices have the potential to accelerate progress by providing a clear framework that fosters interoperability and consistency. The adoption of standardized practices enables compatibility between various manufacturers of charging stations and electric vehicles, ensuring seamless functionality across systems. This not only improves system performance but also significantly enhances the overall user experience. Just as hardware standardization is critical for optimizing user interaction, the standardized use of communication protocols is essential for ensuring compatibility across the entire infrastructure to meet users' needs.

To be effective, these standardized practices must be developed collaboratively, with input from EV manufacturers, charging station producers, operators, and policymakers. Aligning these diverse stakeholders can reduce complexity and drive the industry forward in a cohesive and innovative manner. While collaborative discussions are underway, achieving consensus takes time. The main challenge lies in the fact that the stakeholders are numerous and highly diverse, each with their own agendas and priorities. Additionally, the rapid pace of technological advancement in recent years has outpaced the ability of policymakers to keep up.

Another important discussion point is the dual functionality of charging spots as both parking and charging spaces, which presents a dilemma, particularly in areas where street parking is already limited. As the demand for public EV charging grows, charging stations often occupy valuable parking spaces, creating competition between EV users and drivers of other vehicles. This situation can lead to tension, especially in urban areas where parking scarcity is a persistent issue. The challenge lies in finding a balance between supporting the expansion of EV infrastructure and addressing the broader need for general parking. From an EV user's perspective, a significant issue is the lack of sensors to indicate whether a car is parked at a charging spot or not. Installing sensors at each charging point to detect parked cars would be costly.

One potential solution could involve rethinking parking policies, such as designating specific times for charging-only use or creating more dedicated spaces to serve both purposes. Another suggestion is to strictly forbid parking in EV charging spots, which would address the issue of inaccurate availability information on EV charging services.

Approach

The service design approach was adopted in this project, as it is particularly well-suited for creating comprehensive, user-centered services that not only address user needs but also align with the organization's broader goals. The holistic nature of service design is evident in this project, where a wide range of stakeholders are involved, such as EV drivers, charging point operators, grid operators, car manufacturers, and municipal authorities, and are considered throughout the development process.

In this project, one of the main challenges is balancing user needs with technological and business constraints. For instance, although users may prefer lower costs, charging operators

must consider financial sustainability and ensure that the pricing models support long-term growth. This approach considers all aspects of EV charging, from the technical infrastructure to user interaction. By adopting this approach, the project builds a comprehensive solution that incorporates multiple perspectives rather than focusing solely on individual elements, such as only the mobile application interface.

In the user analysis, participants were selected at random, but included both individual and commercial drivers to ensure a broad demographic representation. However, potential sources of error, such as confirmation bias and situational bias, may still have influenced the results. Confirmation bias refers to the unintentional tendency of researchers to favor information that supports their presumptions. To mitigate this risk, a neutral and open-minded approach was maintained throughout the study. Similarly, situational bias can arise when respondents provide answers influenced by their immediate environment, potentially distorting their overall experience. For example, a respondent might give feedback based on their experience at a specific charging station rather than their overall perception of multiple charging points. Conversely, the goal of interviewing participants while they were actively using public charging infrastructure was to observe their behavior, thoughts, and feelings in the moment of interaction.

Literature study

A literature review was conducted to gain a deeper understanding of EV charging technology, as well as consumer behavior and preferences. The aim of the review was to identify patterns, existing knowledge, and potential information gaps. The literature review delved deeply into various technologies related to charging infrastructure and specific protocols. This approach was intended to advance the project by providing a solid understanding of what was feasible and offering insight into potential issues. Additionally, the literature review helped to identify opportunities within the field.

The project focuses on a Swedish context, but there are many relevant references from the EU that are likely applicable in Sweden, particularly those from the Nordic countries. However, there are a few references from outside the EU that for example discussed ultra-fast chargers improving convenience for EV users during long-distance travel. Since these findings reflect a user perspective on enhanced convenience, they are likely applicable to users in a Swedish context as well.

Research questions

Below is a discussion of the research questions outlined in Chapter 1.2 and how they are addressed.

RQ1: What are the EV drivers' needs for mobile applications linked to public EV charging, and how can these needs be addressed to enhance the user experience?

RQ1 is primarily addressed through the user analysis, which was enriched by insights from the literature review of existing research on user behavior. The user study helped identify key needs among EV drivers, where clustering was used for identifying four interesting themes of feelings, such as powerlessness and confusion. These insights were crucial for understanding the real-world challenges faced by users at public charging stations.

The literature review complemented these findings by offering insights into EV consumer behavior and preferences. Furthermore, the literature review provided frameworks on EV charging technologies and communication protocols, which were essential in determining what technical solutions would be feasible for the design phase. The user needs are addressed by the mockup in the final design and concept development, linking the needs of EV users with apt and feasible technology and functions.

RQ2: How can the company enhance profitability through the mobile application functionalities while aligning with user expectations?

RQ2 is primarily addressed through a deeper understanding of user needs, combined with a thorough market analysis and theoretical framework of charging technologies and communication protocols, with a focus on profitability-enhancing strategies such as energy arbitrage and peak shaving.

Insights gathered from the literature review and user analysis suggest that the company can effectively reach an expanding segment of EV drivers, which has been identified as the chosen target group, and enhance the mobile application with features that appeal to this group. By doing so, the company can attract a broader and more loyal customer base, thereby increasing revenue. Furthermore, by incorporating load management strategies that benefit both the users and the company, such as optimizing charging to reduce costs, the company

can increase profitability and unlock new revenue streams, while expanding its customer base.

8.2 Conclusion

The purpose of this thesis project was to investigate the user needs associated with mobile applications linked to EV charging. The project followed a service design approach and was conducted in collaboration with the company VCG. A combination of literature review, user analysis, and market analysis was carried out to gather insights and better understand the demands and challenges EV users face when interacting with mobile applications.

To address the research questions, a comprehensive user analysis was carried out, employing a service design approach, focused on identifying needs, behavior, and existing problems. Key findings revealed that EV users face challenges such as lack of transparency, limited access to support, deficient filtering options, and the need to use multiple applications. These problems lead to users not trusting the infrastructure. Moreover, the price of public charging is a significant factor in selecting charging locations, with many cost-sensitive users prioritizing affordability, particularly those without access to home chargers. The study led to seven key insights that were delivered to VCG. The insights were addressed with solutions that were implemented in a mockup.

The conclusion of this project is that the implementation of the final design of the mockup addresses the identified challenges and enhances the user experience.

8.3 Recommendation and future work

To move forward with the development of the mobile application, there is still a lot of work to be done. Real-world stakeholder tests of the mockup are needed to gather additional insights and iterate on the design before building the actual prototype and backend systems for functionality. An additional part of the service extension is the new brand extension, and the actions presented should be implemented for the service's success.

At a societal level, there are some key tangible suggestions for a better solution that should be investigated further. The suggestions are compiled in the list below:

- There should be a consensus on which parking rules apply to EV charging locations nationally. Currently, inconsistent laws create high levels of uncertainty and confusion

regarding charge point availability, leading to distrust in the public charging infrastructure.

- There should be a national open database or system that houses all the different charge locations and real-time data related to each charge point. Currently, users have to switch between multiple different mobile applications just to navigate the public charging space. The idea is that the open database can facilitate communication between different networks and systems, similar to how email or SMS can be sent between different networks.
- There should be standardized laws regarding which protocols are used by charge point operators, charge point manufacturers, and EV manufacturers. Currently, users face interoperability issues between vehicles and charge points, which creates frustration and resentment towards the infrastructure.

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APPENDIX A - Distribution of protocols in the network.

Figure A presents an illustration of the communication channels and their relation to the different actors, as well as the various protocols among the different actors (Klapwijk & Driessen, 2017).

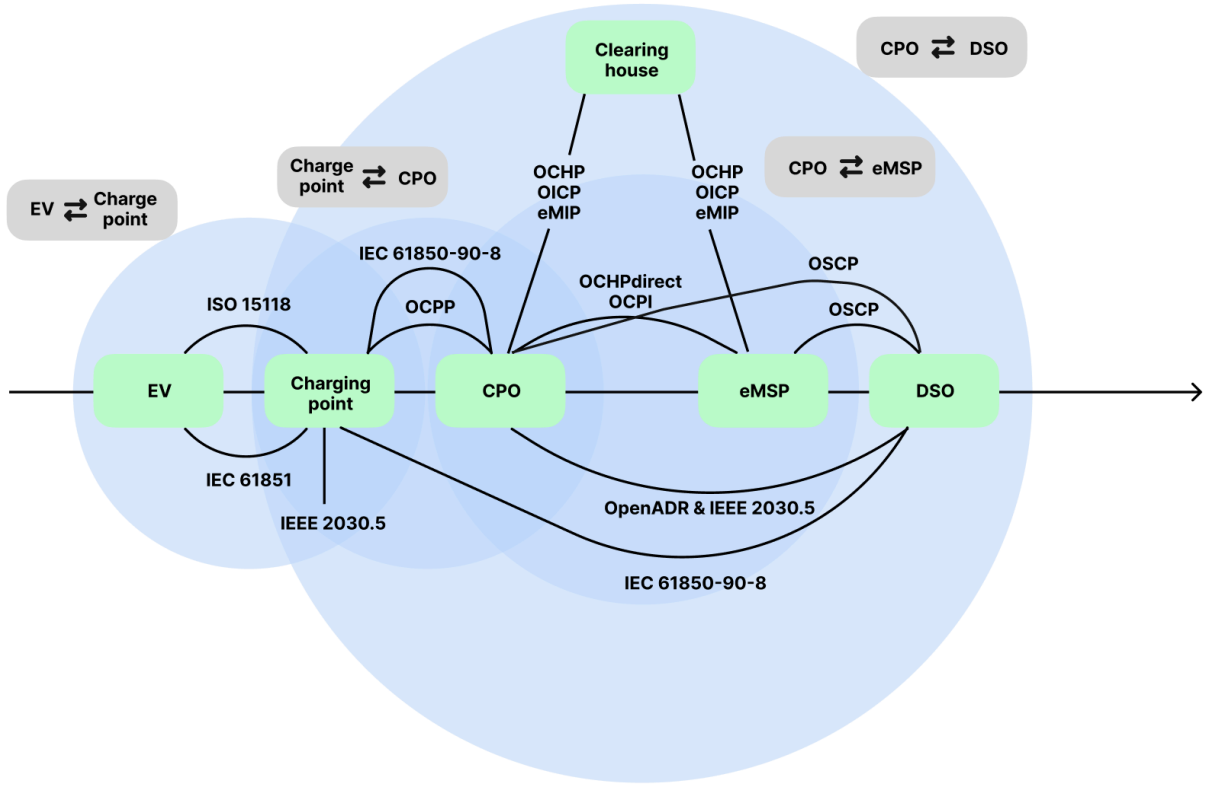


Figure A. Illustration of the communication channels and their relation to the different actors.

APPENDIX B - Interview guide

The interview guide used during pop-ups. The questions composed the framework of the interview, and follow-up questions were always asked.

SHORT INTERVIEW

Introduktion:

Hej, vi är studenter från KTH som utvecklar en app för elbilsladdning. Vi vill veta lite mer om användarbehov och dina tankar kring appar för elbilsladdning.

Namn: **Kön & Ålder (uppskattad):**

1. Använder du elbil idag?
   Om ja - vilken elbil har du?
   Om nej- ok, tack för idag!

2. Hur ofta brukar du använda publika laddstationer?

3. Hur prioriterar du faktorer som laddningshastighet, kostnad och läge när du väljer en laddningsstation?

4. Hur hittade du hit idag? Vilken APP? **Varför den appen?**

5. Vad är din upplevelse av att använda appar kopplade till publika laddstationer? Vad är bra? Vad är dåligt?

6. Är det något du önskar du kunde göra i appen? Vilka funktioner saknar du?

7. Vad har du känt för frustration över om elbilsladdning apparna? (Jobbigt, lång tid)

8. När du planerar en längre resa, hur brukar du tänka/göra för att ladda bilen tillräckligt?

9. Hur upplever du betalningarna när du laddar? Hur kan man förbättra betalningsmetoder/system?

Stort tack för att du ställde upp! Ha en fortsatt trevlig dag!

APPENDIX C - Overview of mobile application comparison

Figure C shows a blurred overview of the full comparison of the different apps, mapped in *Figma*.



Figure C. Blurred out preview of mobile application comparison.

APPENDIX D - Original quote and translation

Table D displays the quotes originating from the interviews, either spoken or written, and are extracted from the CJM or the clustering.

Table D. Original quotes translated from Swedish to English.

Original quote:	**Used translation:**
"Det är vanligtvis upptaget, och du måste ta hänsyn till avståndet om du kör med få kilometer kvar."	"It's usually occupied, and you need to account for the distance if you're driving with not that many kilometers left." - Respondent P17
"Var tvungen att ringa support eftersom jag inte kunde stoppa laddningen via appen och inte kunna ta bort laddaren."	"Had to call support because I couldn't stop the charging via the app, and couldn't remove the charger." - Respondent P15
"Det är okej, jag kan ladda överallt"	"It's okay, I can charge everywhere" - Respondent P7
"[...] men det finns appar som inte startar BankID direkt, vilket innebär att du måste avsluta appen, hitta BankID, identifiera dig och sedan återgå till appen."	"[...] however, there are apps that don't launch BankID directly, which means you need to exit the app, find BankID, identify yourself, and then return to the app." - Respondent P14
" [...] ibland känner jag att det är för mycket funktioner på apparna så man blir ju nästan förvirrad"	"[...] sometimes I feel that there are too many features in the apps, so you almost become confused." - Respondent P3
"Det fungerar bra, jag har allt jag behöver"	"It works fine, I have all the things I need"- Respondent P5
"Det är på tok för många appar. Vissa funkar inte, vissa funkar bättre. Oavsett är det en bökig, svårnavigerad djungel."	"There are way too many apps. Some don't work, some work better. Regardless, it's a messy, hard-to-navigate jungle." - Respondent P3
"Jag skulle vilja ha en app för alla [offentliga laddpunkter], kanske finns det en, men det känner jag inte till i alla fall"	"I would like to have an app for all [public charge points], maybe there is but I don't know about it anyway" - Respondent P9
"Jag hade en KIA förut och den hade alltid problem att disconnecta själva kabeln från	"I had a KIA before and it always had problems disconnecting the cable itself from

bilen efter att jag hade laddad den full. Var tvungen att operatören och be dem koppla loss mig hela tiden.”	the car after I had fully charged it. Had to call the operator and ask them to disconnect me all the time.” - Respondent P1
“ [...] man kan inte lita på att det alltid finns plats när man kommer fram, eller att dom fungerar som dom ska för den delen.”	" [...] you can't trust that there is always room when you arrive, or that they work as they should for that matter." - Respondent P11
"Shit, här fanns ingen kabel, ska kolla om jag har en, två sek [börjar leta i bagageutrymmet efter en laddkabel] [inser att hon inte har rätt kabel med sig], aja får köra vidare”	"Shit, there wasn't a cable here, I'll check if I have one, [starts looking in the trunk for a charging cable] [realizes she doesn't have the right cable with her], I’ll have to drive on" - Respondent P12
“Jag försöker alltid hitta den billigaste laddaren i närheten”	“I always try to find the cheapest charger nearby” - Respondent P11

APPENDIX E - CJM

Figure E.1 shows an overview of the CJM that was carried out.

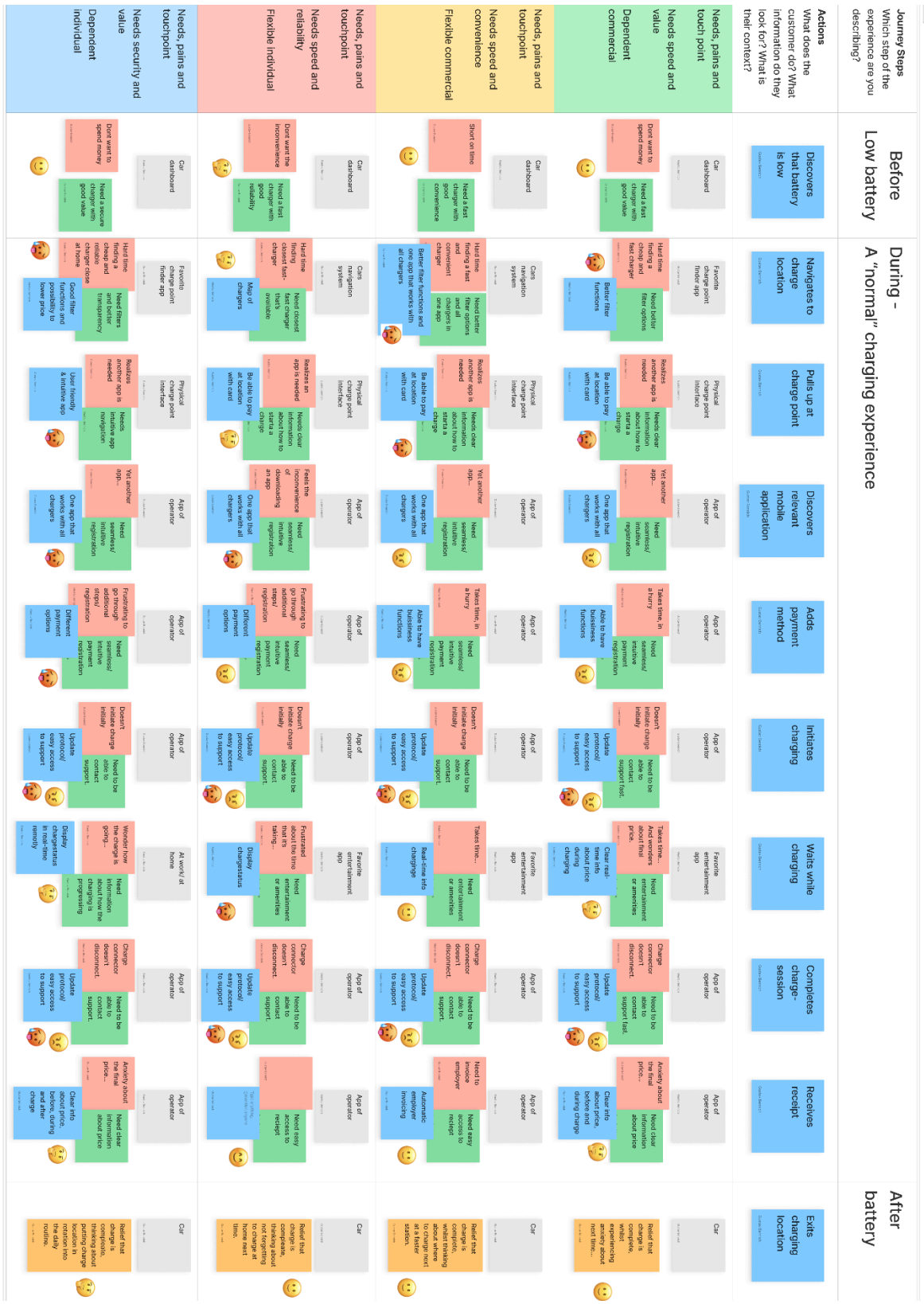


Figure E.1. Customer Journey Map.

Figure E.2 presents the explanation of color grading of the sticky notes, and the emotions expressed with the emojis.

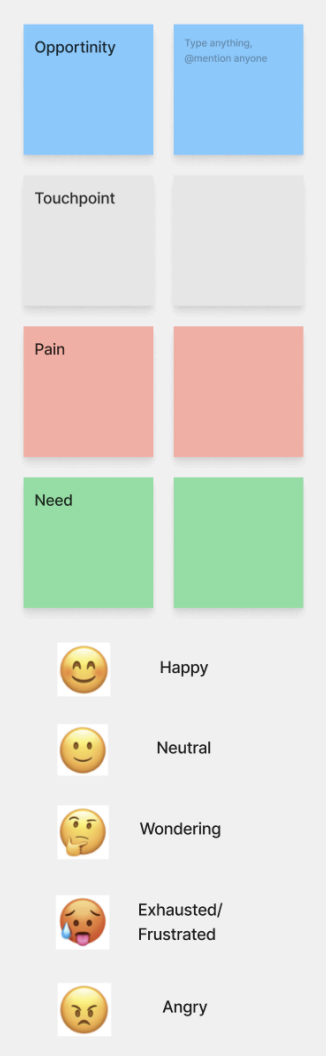


Figure E.2. Explanation of colors and emojis in CJM.

APPENDIX F - Extraction of ideation

Figure F. presents an extraction of the cluster of ideas from the ideation phase displayed on an open canvas or digital whiteboard to easily sort and cluster different ideas.



Figure F. Overview of ideas from the ideation phase.

APPENDIX G - Overview of layout of mockup

Figure G shows an overview of the layout of the mockup defined in concept development.

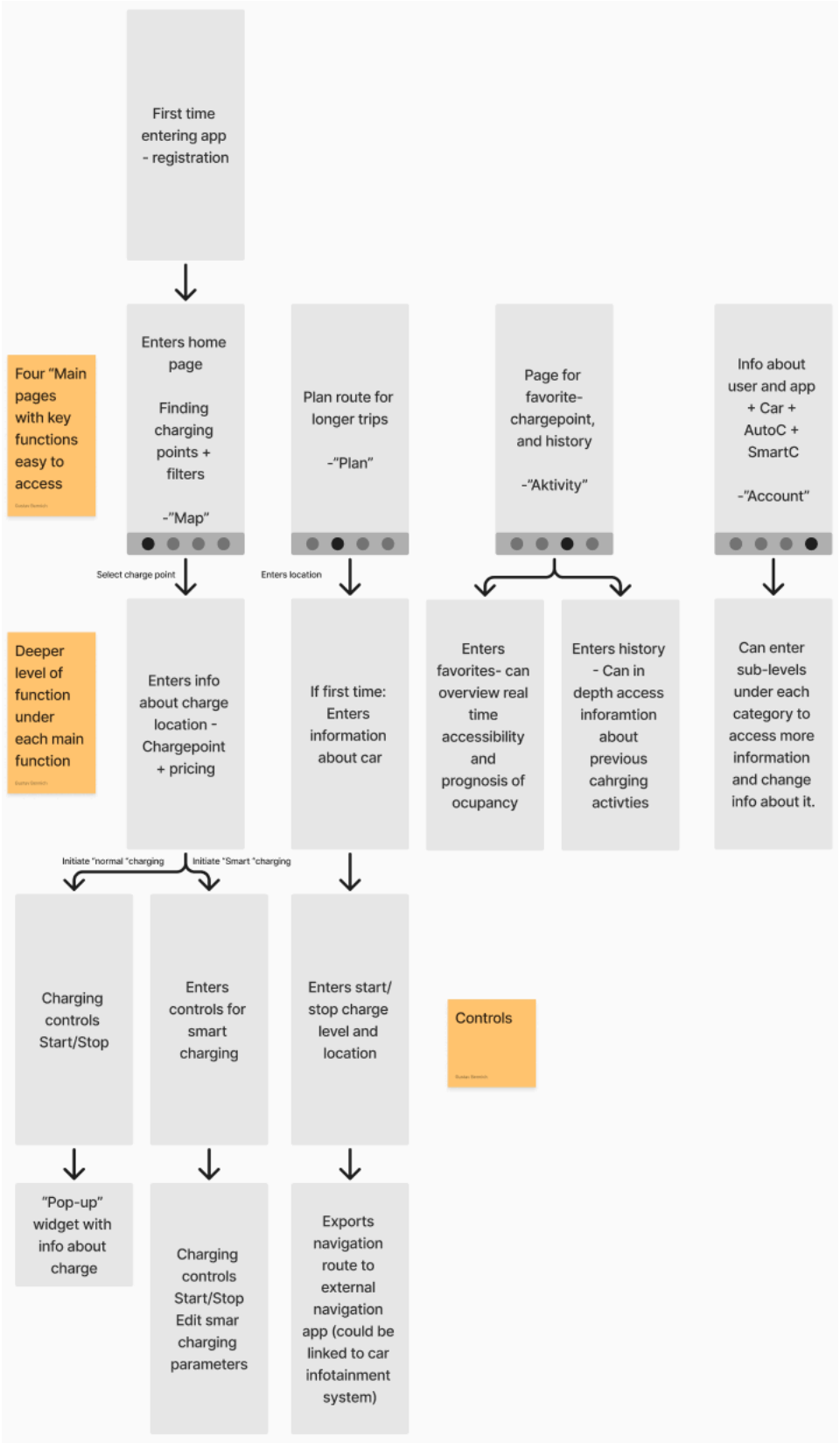


Figure G. Overview of layout.

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